

16CS53
Computer Graphics
and
Visualization

Department of
Computer Science

presentation
notes

Unit 4

Viewing

By
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Objectives

Unit 4

Viewing

- Introduce the classical views
- Learn the benefits and drawbacks of each type of view.
- Introduce the mathematics of projection
- Introduce OpenGL viewing functions
- Look at alternate viewing APIs
- Derive the projection matrices used for standard OpenGL projections
- Introduce oblique projections
- Introduce projection normalization

Introduction

Unit 4

Viewing

- Discussion about the types of views that we can create and why we need more than one type of view.
- How an application program can specify a particular view within OpenGL.
- Viewing process has 2 parts
 - Use of **projection matrix** to switch vertex representations from the object frame to the eye frame.
 - Type of preferred projection (**parallel or perspective**).

Classical Viewing

Unit 4

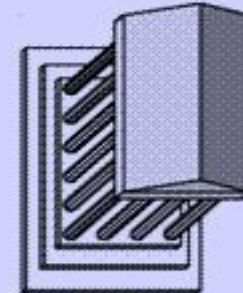
Viewing



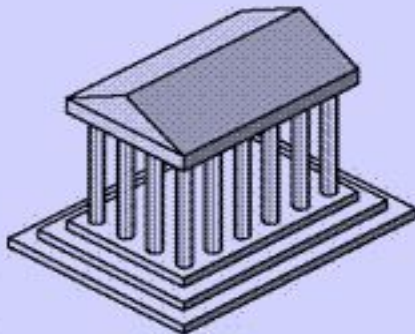
Front elevation



Elevation oblique



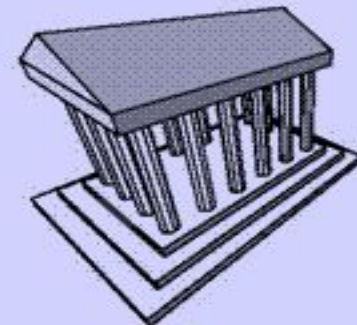
Plan oblique



Isometric



One-point perspective



Three-point perspective

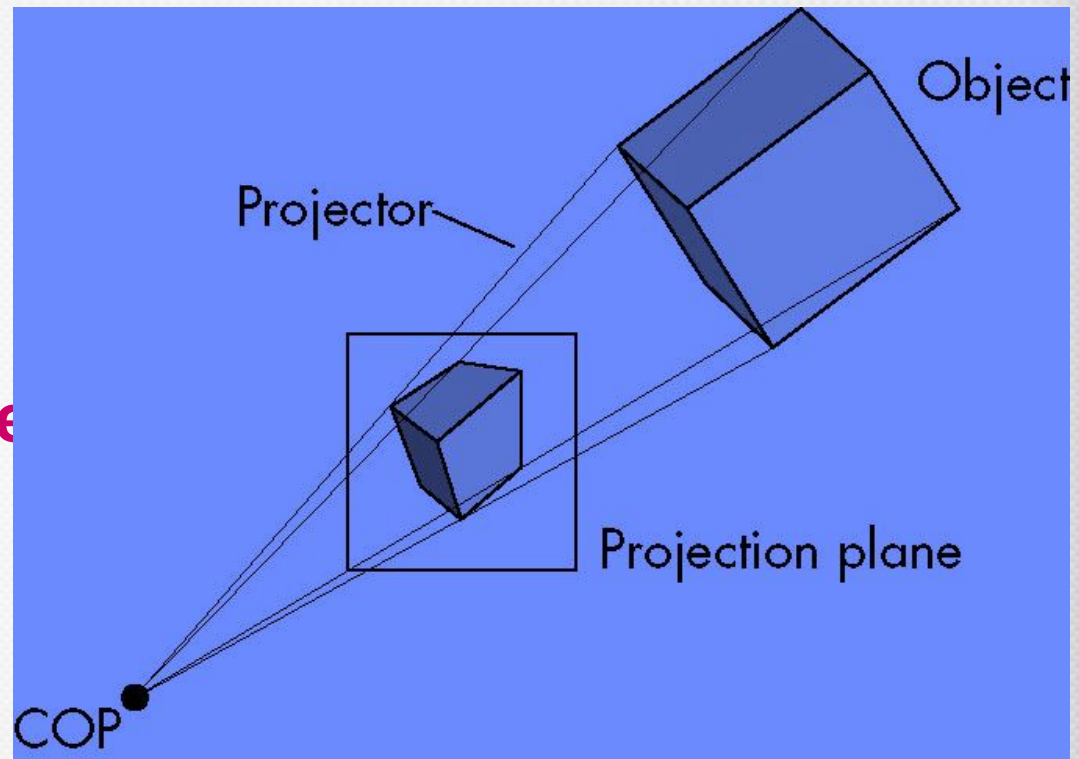
Classical Viewing

Unit 4

Viewing

Elements of classical viewing

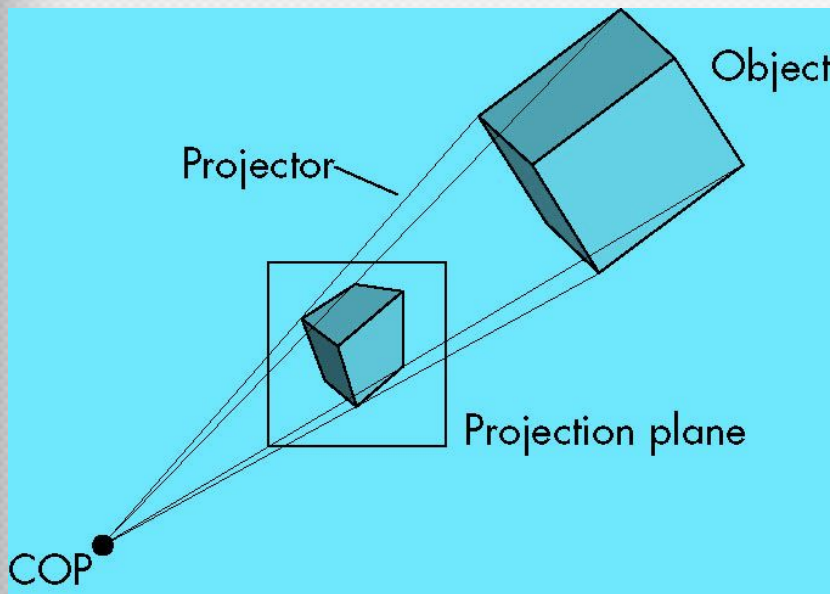
- Objects
- A viewer
- Projectors
- A projection plane



Classical Viewing

Unit 4

Viewing



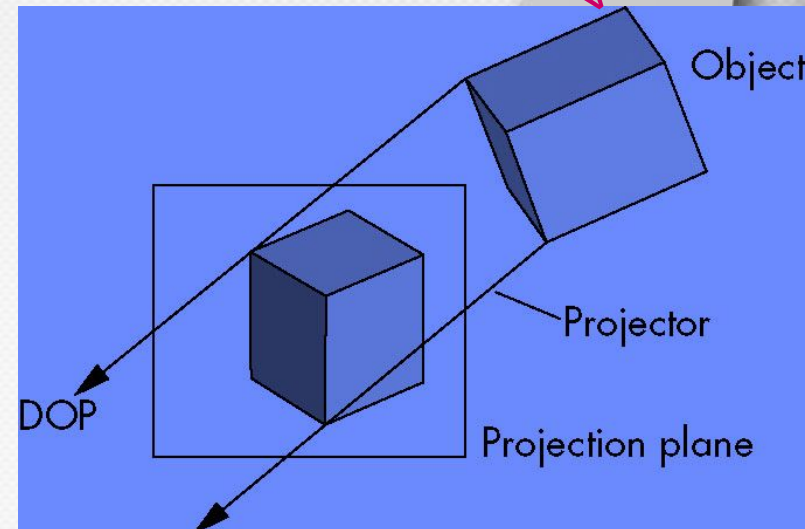
- The projectors meet at the centre of projection.
- COP corresponds to
 - **Centre of the lens** *in the camera.*
 - **Origin of the camera frame** for perspective views *in the graphics system.*
- Projection surface is plane.
- Projectors are straight lines.

Classical Viewing

Unit 4

Viewing

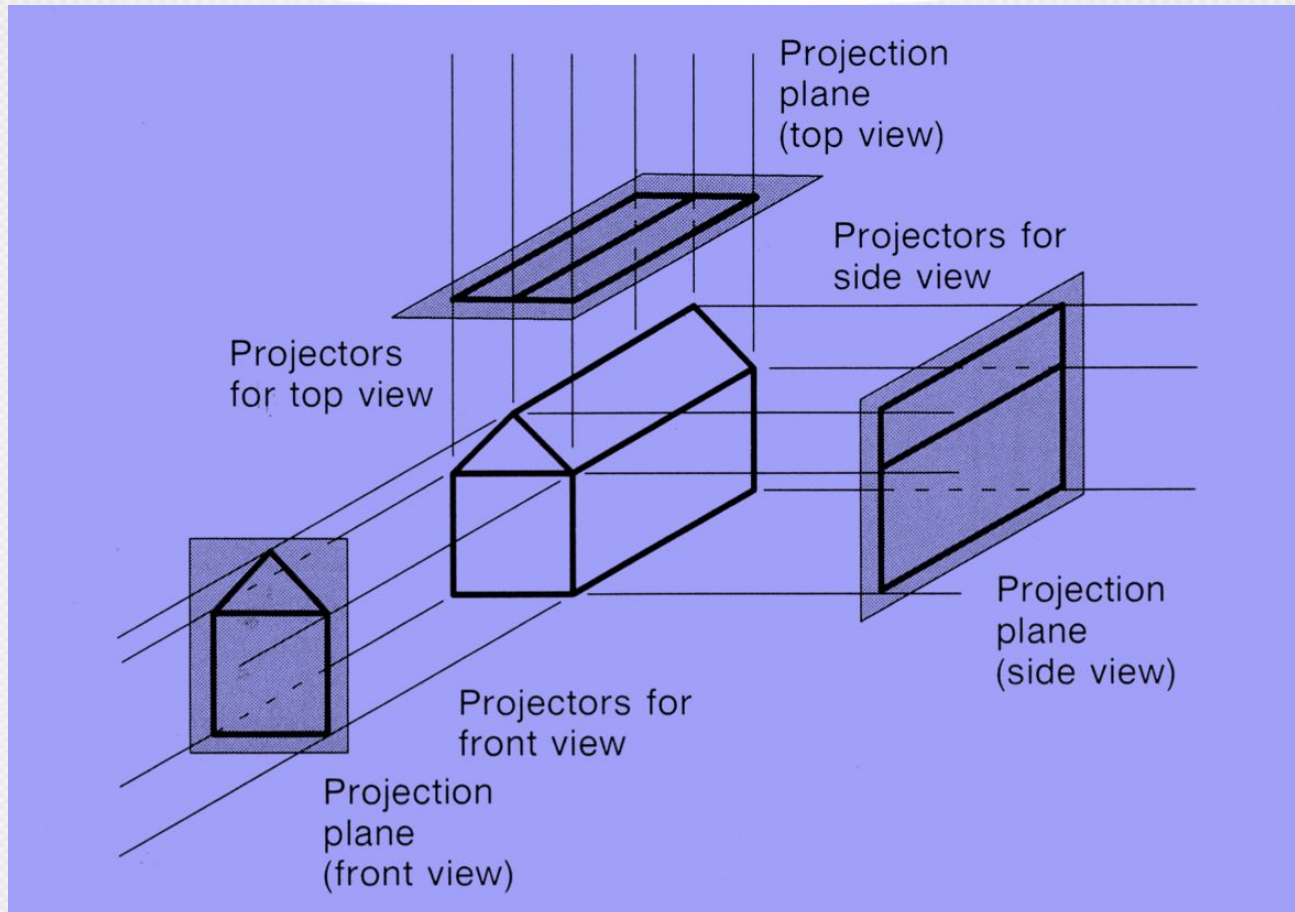
- As COP moves to infinity, the projectors become parallel and COP is replaced by DOP (Direction of Projection).
- Size of the image remains as that of the object even though COP is ∞ ly far from the objects.
- **Perspective views** : views within finite COP.
- **Parallel views** : views with infinite COP. For parallel views, the origin of camera frame lies in the projection plane.



Classical Viewing

Unit 4

Viewing



Classical Viewing

Unit 4

Viewing

- Viewing requires three basic elements
 - One or more objects
 - A viewer with a projection surface
 - Projectors that go from the object(s) to the projection surface
- Classical views are based on the relationship among these elements.
- The viewer picks up the object and orients it how she would like to see it.
- Each object is assumed to be constructed from **principal faces**.

Unit 4

Viewing

Types of Classical Viewing

Classical Viewing Projections

Unit 4

Viewing

Perspective
(Converging Projection)

Parallel
(Parallel Projection)

One point

Two point

Three point

Orthographic
Projectors are $\perp$ to
view plane

Oblique
Projectors are not $\perp$
to view plane

Front

Top

Side

Multiview

View plane $\parallel$ to
principal plane

Isometric

Dimetric

Trimetric

Axonometric

View plane not $\parallel$ to
principal plane

Classical Viewing Projections

Unit 4

Viewing

Perspective

(Converging Projection)

One point

Two point

Three point

Parallel

(Parallel Projection)

Orthographic

Projectors are $\perp$ to view plane

Oblique

Projectors are not $\perp$ to view plane

Front

Top

Side

Multiview

View plane $\parallel$ to principal plane

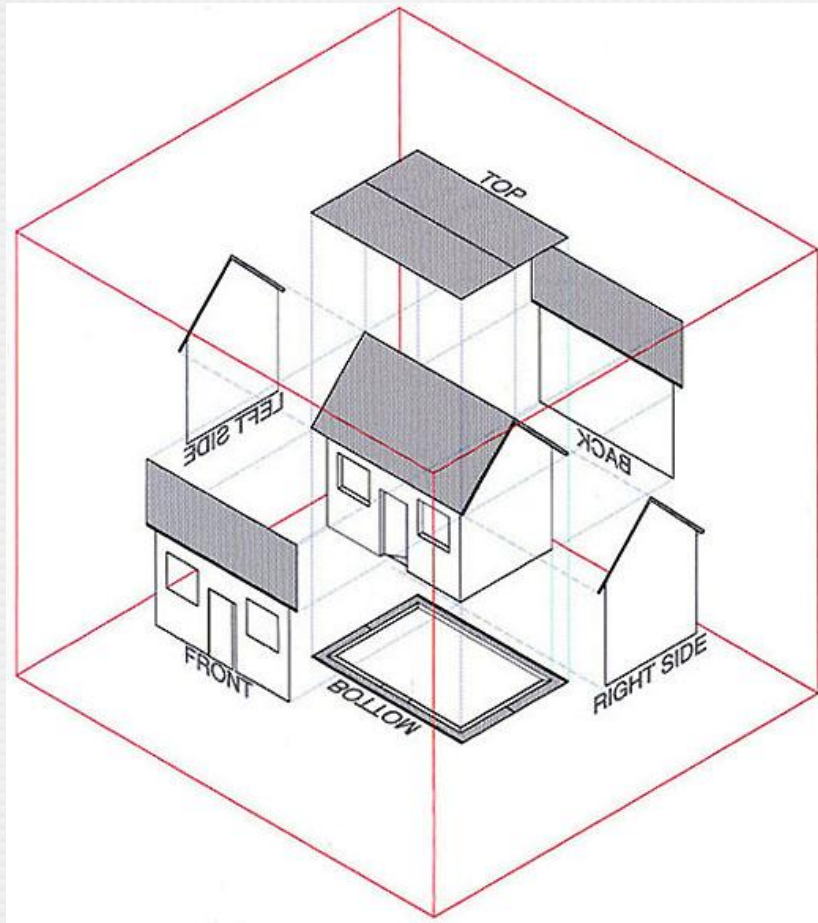
Isometric

Dimetric

Trimetric

Axonometric

View plane not $\parallel$ to principal plane



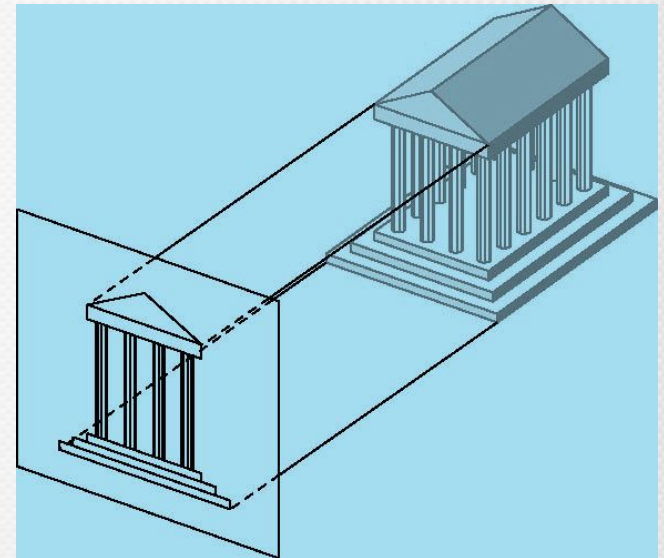
Orthographic Projection

Orthographic Projection

Unit 4

Viewing

- Projectors are **perpendicular** to the projection plane.
- Usually 3 views are used in an orthographic projection to display the objects.
 - Front view
 - Top view
 - Side view



Orthographic Projection

Unit 4

Viewing

Advantages

- Preserves both distances and angles.
- Since there is no distortion, well suited for working drawings.

Types

- **Multiview** orthographic projection
- **Axonometric** orthographic projection

Classical Viewing Projections

Unit 4

Viewing

Perspective

(Converging Projection)

One point

Two point

Three point

Parallel

(Parallel Projection)

Orthographic

Projectors are $\perp$ to view plane

Oblique

Projectors are not $\perp$ to view plane

Front

Top

Side

Multiview

View plane $\parallel$ to principal plane

Isometric

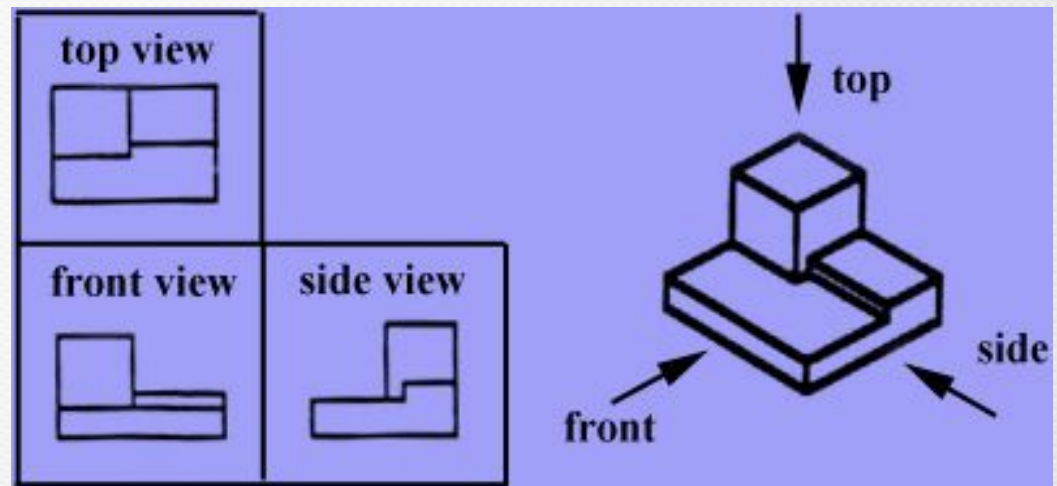
Dimetric

Trimetric

Axonometric

View plane not $\parallel$ to principal plane

Multiview Orthographic Projection

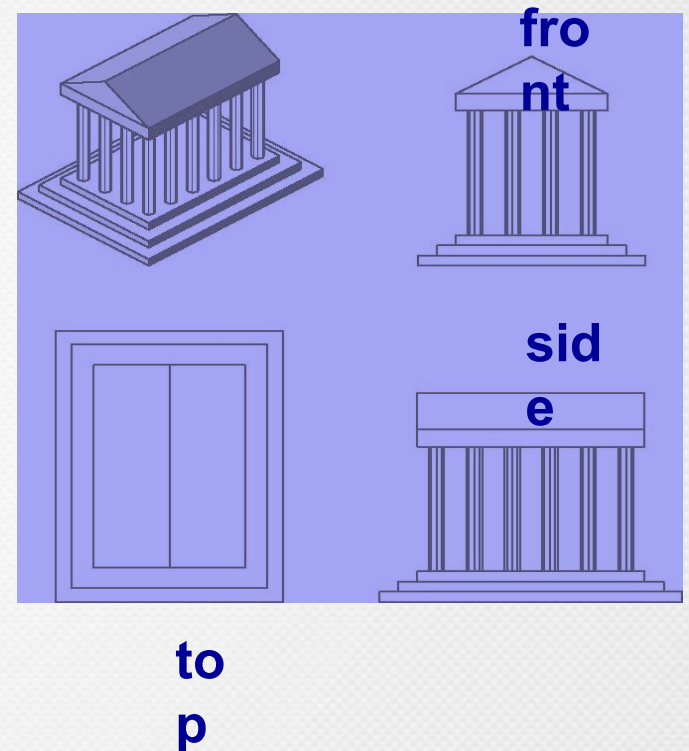


Multiview Orthographic Projection

Unit 4

Viewing

- **Projection plane is parallel to principal face.**
- Viewer usually needs at least two views to visualize how an object looks like.
- Usually form front, top, side views.

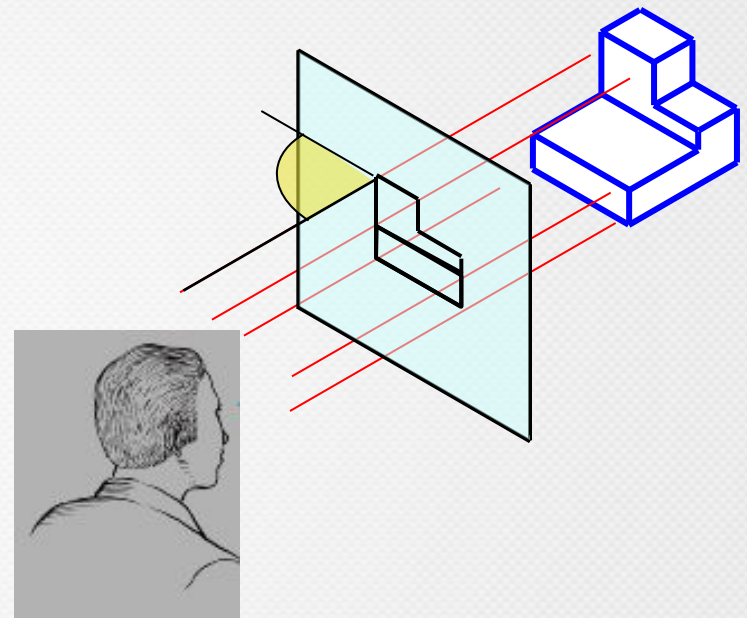


Multiview Orthographic Projection

Unit 4

Viewing

- The observer is considered to be at an infinite distance.
- **Projecting lines**
 - parallel to one another.
 - perpendicular to the projection plane.

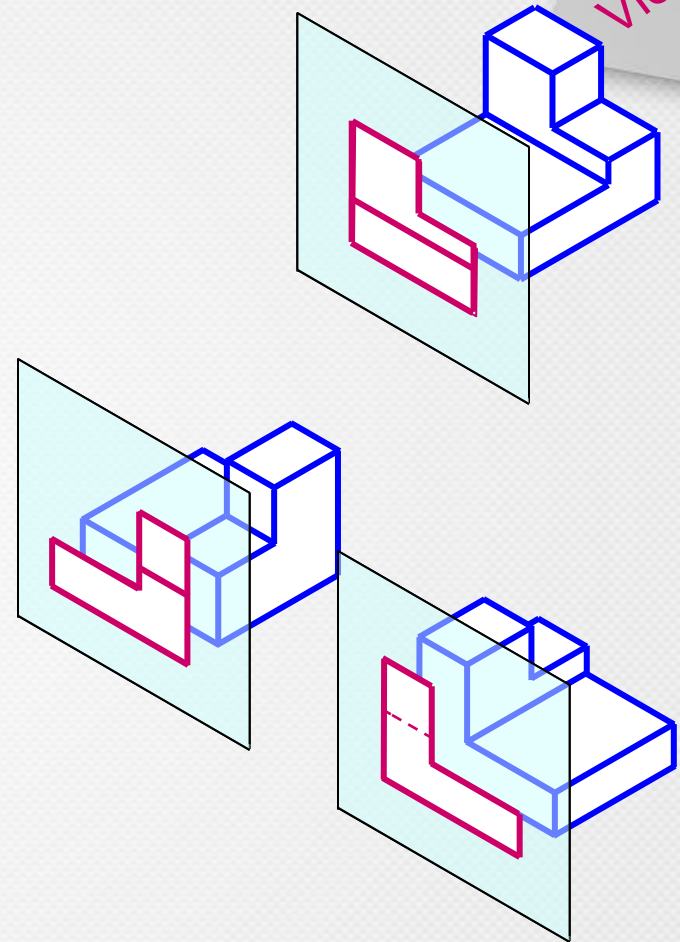


Multiview Orthographic Projection

Unit 4

Viewing

- One of the faces of the object to be drawn is **placed parallel to the projection plane**.
- The observer can only see one side at a time.
- That is why, in most cases, one must draw several views of the object to portray it completely.

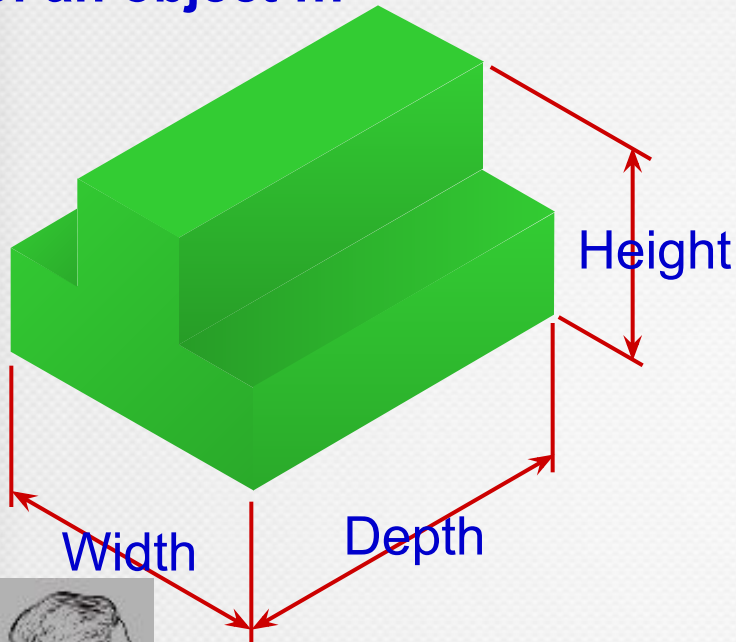


Multiview Orthographic Projection

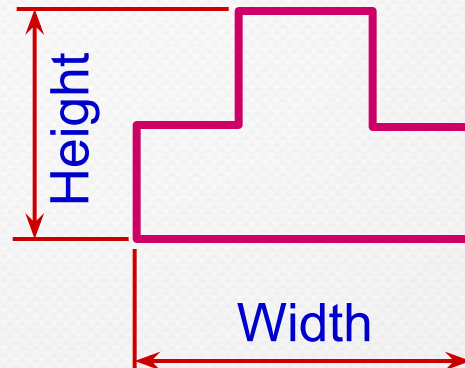
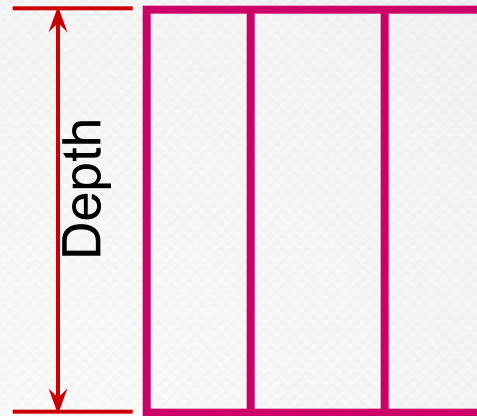
Unit 4

Viewing

Three principle dimensions
of an object ...



... can be presented only
two in each view.



Multiview Orthographic Projection

Unit 4

Viewing

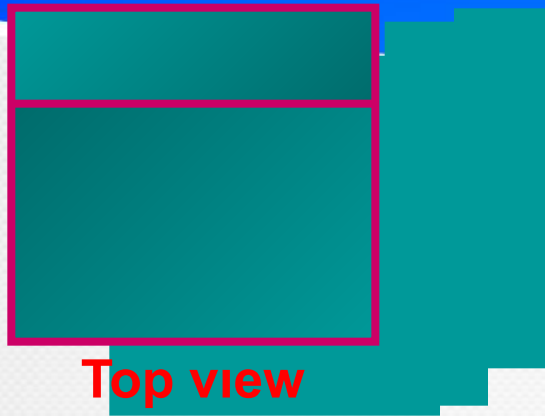
To Obtain Multiview Representation of an Object

1. Revolve the **object** with respect to **observer**.
2. The **observer** move around the **object**.

Revolve the Object

Unit 4

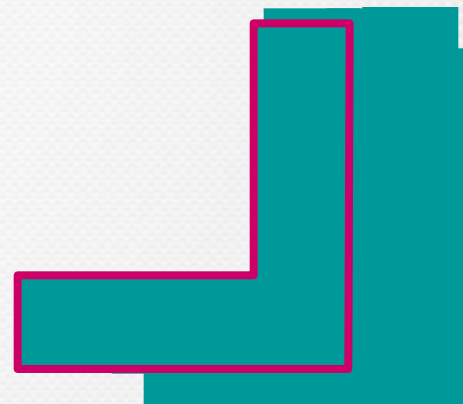
Viewing



Top view



Front view



Right side view

Multiview Orthographic Projection

Unit 4

Viewing

Top view



Observer
Move Around



Front view



Right side view

OpenG

Multiview Orthographic Projection

Unit 4

Viewing

Advantages

- Preserves both distances and angles.
- Shapes preserved.
- Can be used for measurements in Building plans.

Disadvantage

- Cannot see what object really looks like because many surfaces hidden from view.

Classical Viewing Projections

Unit 4

Viewing

Perspective

(Converging Projection)

One point

Two point

Three point

Parallel

(Parallel Projection)

Orthographic

Projectors are $\perp$ to view plane

Oblique

Projectors are not $\perp$ to view plane

Front

Top

Side

Multiview

View plane $\parallel$ to principal plane

Isometric

Dimetric

Trimetric

Axonometric

View plane not $\parallel$ to principal plane

Axonometric Projections

Unit 4

Viewing

4 BHK Axonometric View

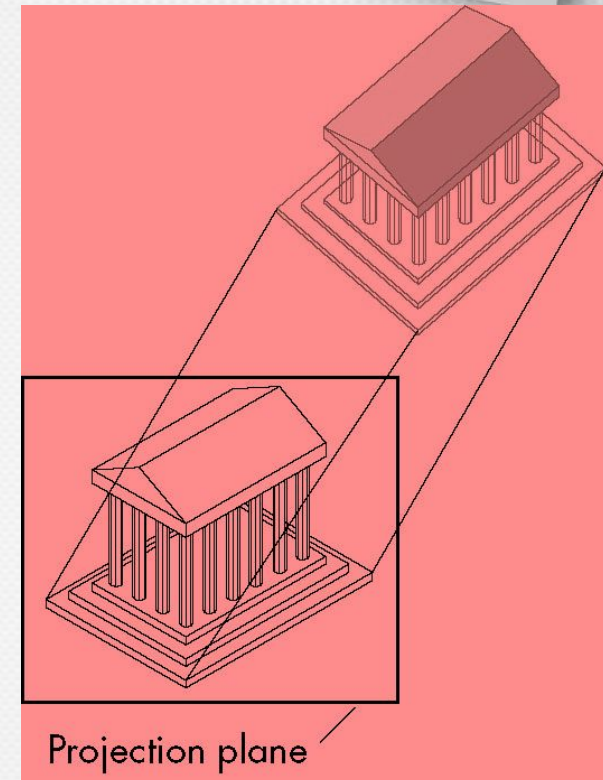


Axonometric Projections

Unit 4

Viewing

- The projectors are still orthogonal to the projection plane, but the **projection plane can have any orientation with respect to the object.**
- Used to create a pictorial drawing of an object, where the object is rotated along one or more of its axes relative to the plane of projection.

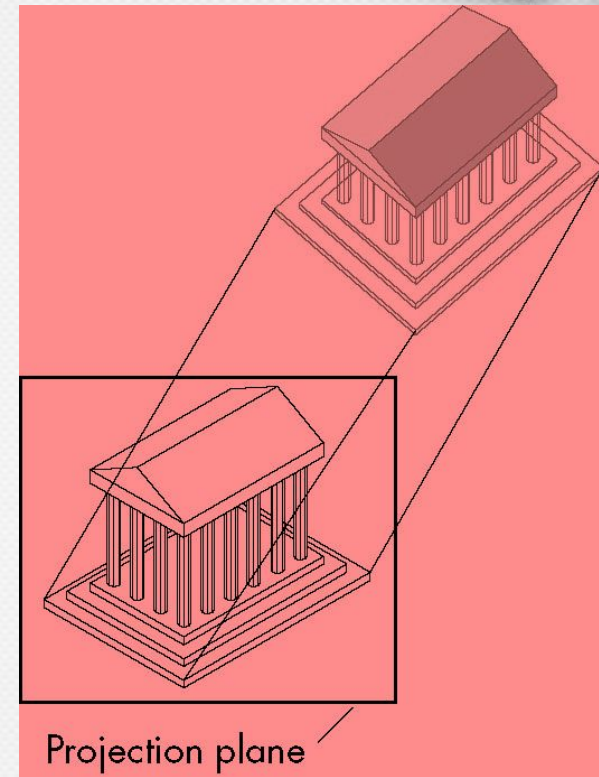


Axonometric Projections

Unit 4

Viewing

- Types of axonometric projection
 - Dimetric
 - Trimetric
 - Isometric
- Used extensively in architectural and mechanical design.



Axonometric Projections

Unit 4

Viewing



Classical Viewing Projections

Unit 4

Viewing

Perspective

(Converging Projection)

One point

Two point

Three point

Parallel

(Parallel Projection)

Orthographic

Projectors are $\perp$ to view plane

Oblique

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Top

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Multiview

Viewplane $\parallel$ to principal plane

Isometric

Dimetric

Trimetric

Axonometric

Viewplane not $\parallel$ to principal plane

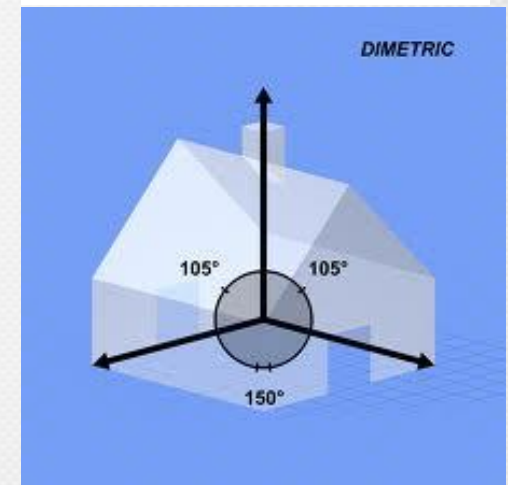
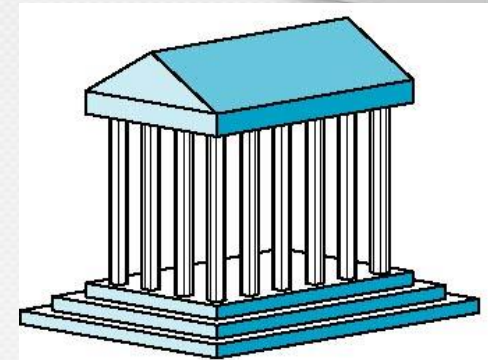
Types of Axonometric Projections

Unit 4

Viewing

Dimetric projection

- Projection plane is placed symmetrically with respect to two of the principal faces.
- The angles between the projection of the axes in dimetric projection renders **two of the three to be equal**.
- To draw the outline of an object in dimetric projection, two scales are required.



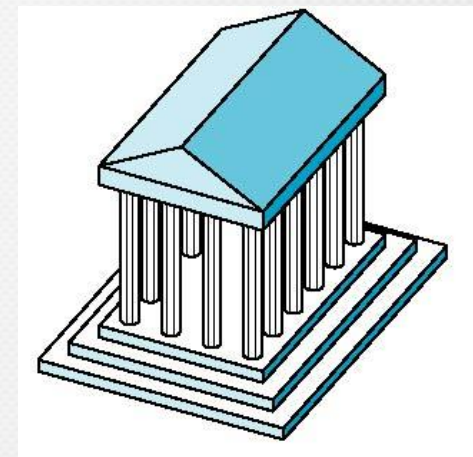
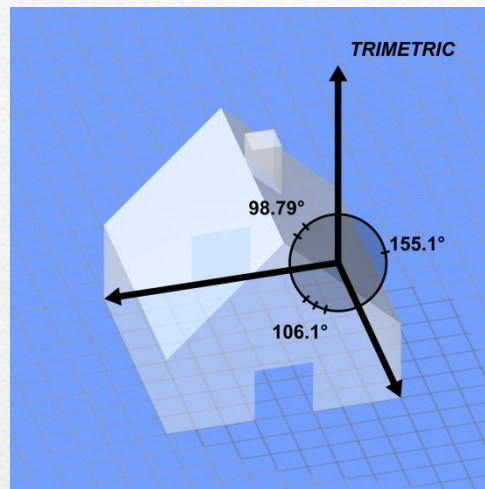
Types of Axonometric Projections

Unit 4

Viewing

Trimetric projection

- In trimetric projection, the projection of the three angles between the axes are unequal.
- Thus, three separate scales are needed to generate a trimetric projection of an object.



Unit 4

Viewing

Isometric Projections



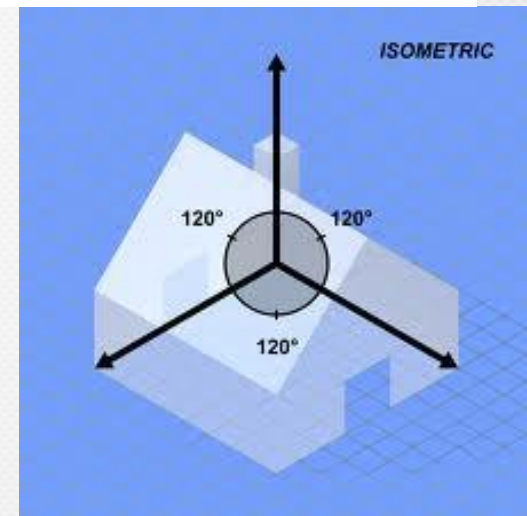
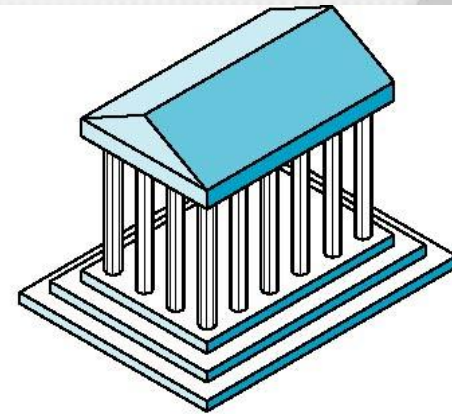
Types of Axonometric Projections

Unit 4

Viewing

Isometric projection

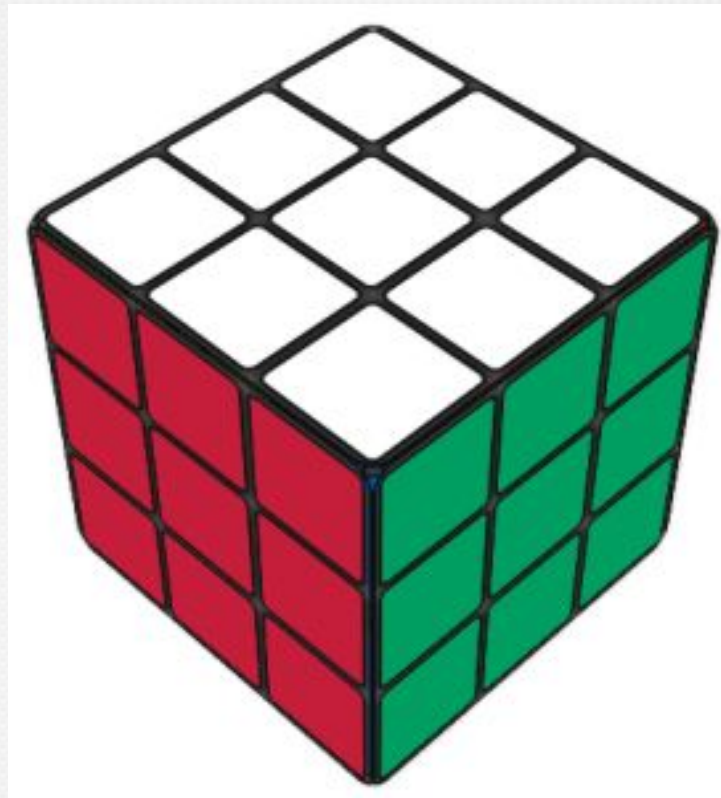
- In isometric projection the angles between the projection of the axes are equal i.e, 120° .



Cube Isometric View

Unit 4

Viewing



Axonometric Projections

Unit 4

Viewing

Advantages and Disadvantages

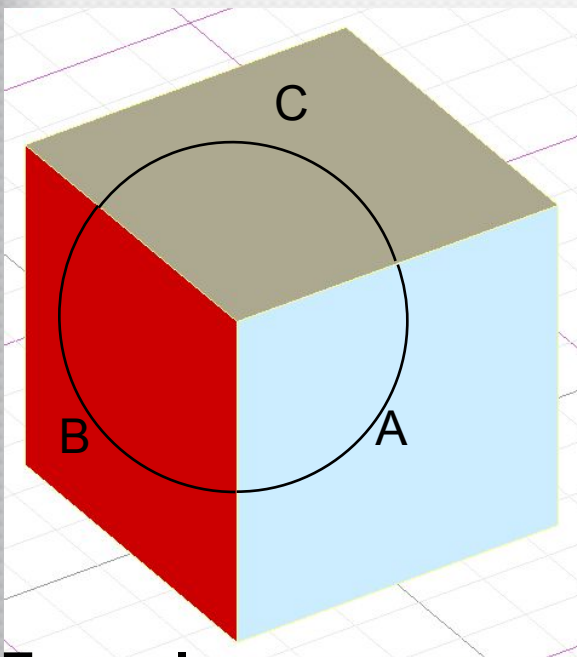
- Lines preserved but angles are not.
- Can see three principal faces of a box-like object.
- Some optical illusions are possible - Parallel lines appear to diverge.
- Does not look real because far objects are scaled the same as near objects.
- Used in CAD applications.

Axonometric Projections

Unit 4

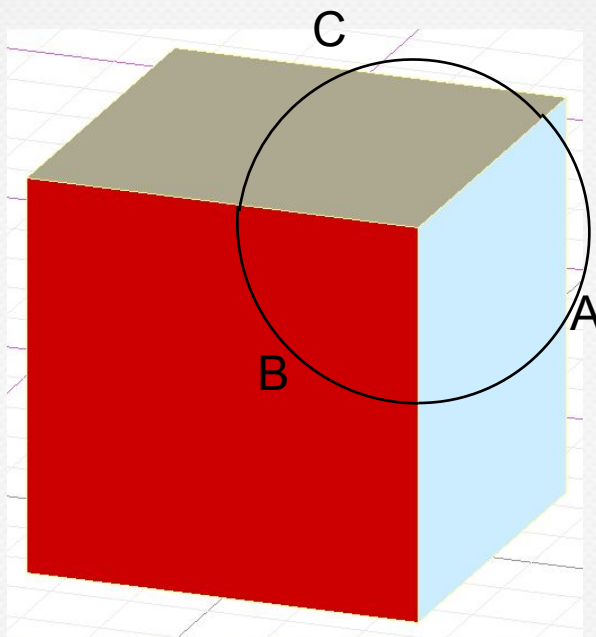
Viewing

TRIMETRIC



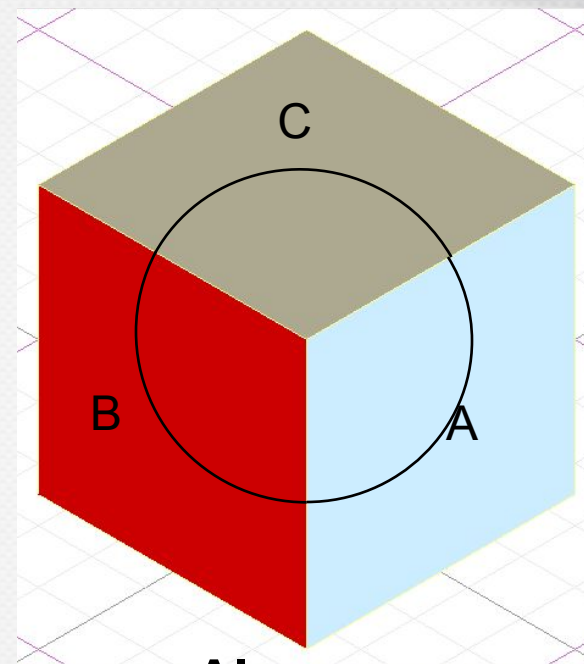
Example:
 $A=120^\circ$ $B=130^\circ$ $C=110^\circ$

DIMETRIC



Example:
 $A=C=131.5^\circ$ $B=97^\circ$

ISOMETRIC



Always:
 $A = B = C = 120^\circ$

Classical Viewing Projections

Unit 4

Viewing

Perspective

(Converging Projection)

One point

Two point

Three point

Parallel

(Parallel Projection)

Orthographic

Projectors are $\perp$ to view plane

Oblique

Projectors are not $\perp$ to view plane

Front

Top

Side

Multiview

Viewplane $\parallel$ to principal plane

Isometric

Dimetric

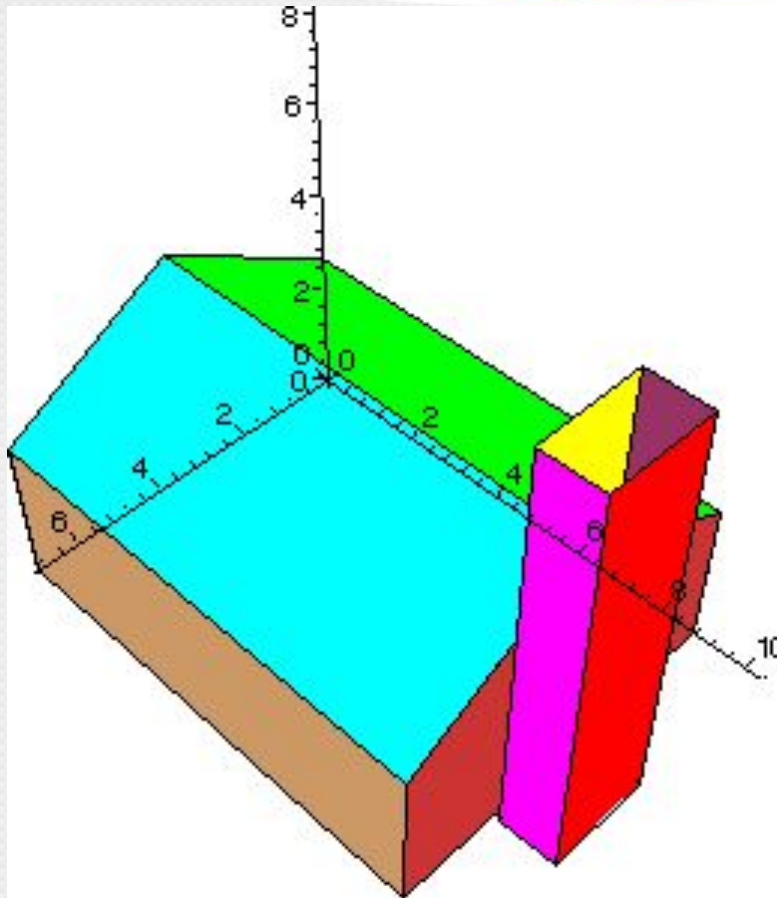
Trimetric

Axonometric

Viewplane not $\parallel$ to principal plane

Unit 4

Viewing



Oblique Projections

Oblique Projections

Unit 4

Viewing

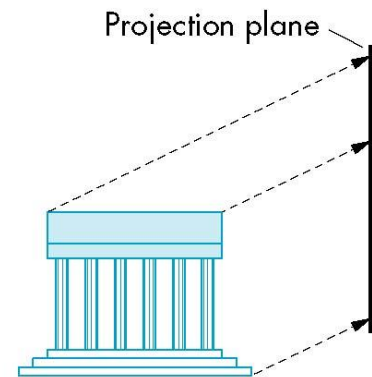
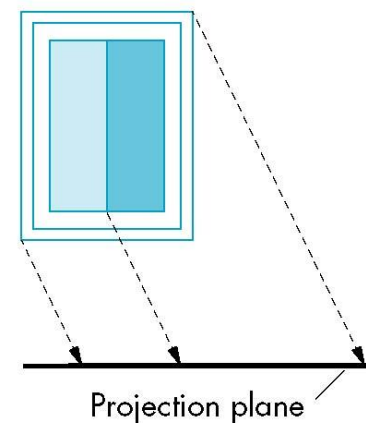
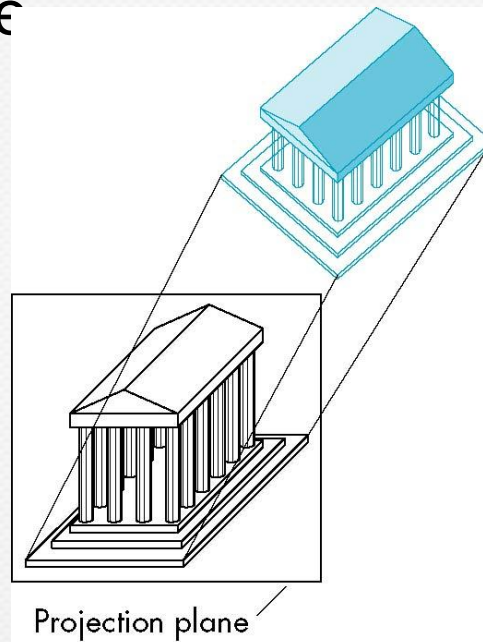
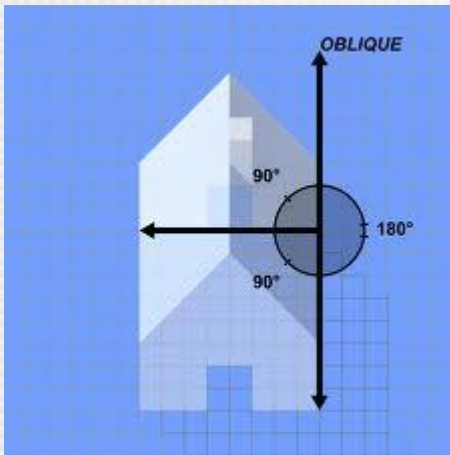
- In oblique projections, the **projectors are parallel** to each other but are **not perpendicular to the plane of projection.**
- One of the object's principal faces is parallel to the plane of projection.

Oblique Projections

Unit 4

Viewing

- One of the most general parallel views.
- Oblique projections are obtained by allowing the projectors to make an **arbitrary angle** with the projection plane

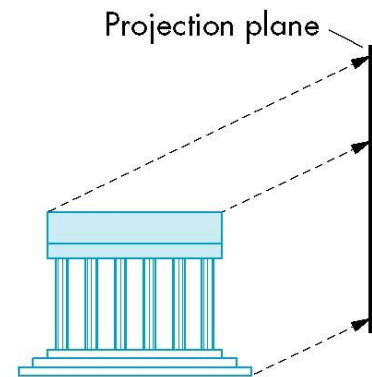
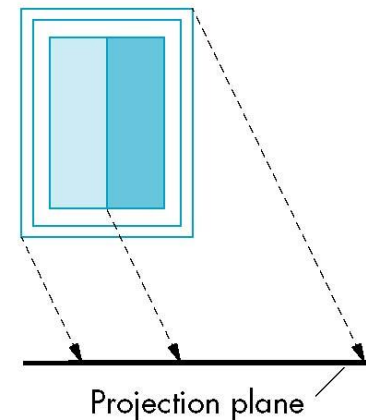
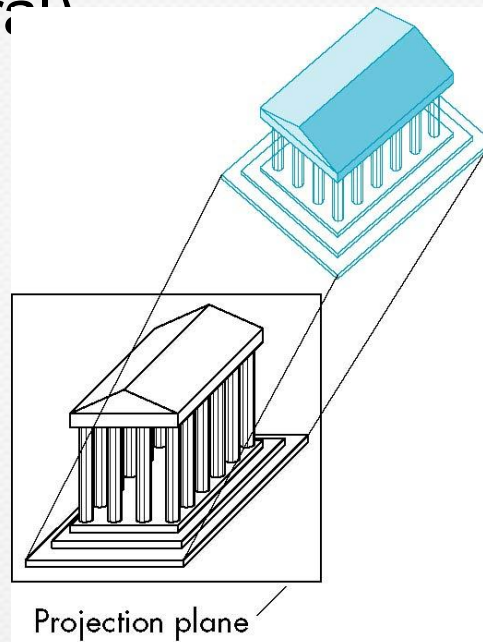
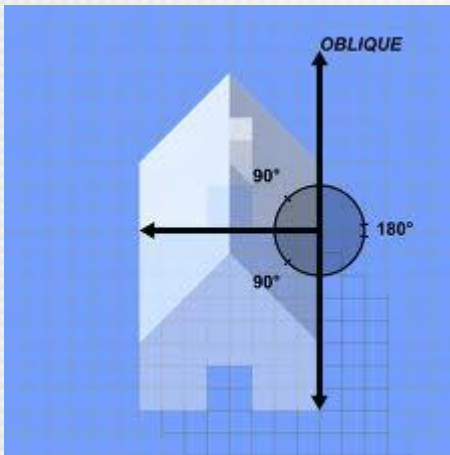


Oblique Projections

Unit 4

Viewing

- Angles of the objects face that are parallel to the projection plane are preserved.
- They are the most difficult views to construct by hand. (Unnatural)



Unit 4

Viewing



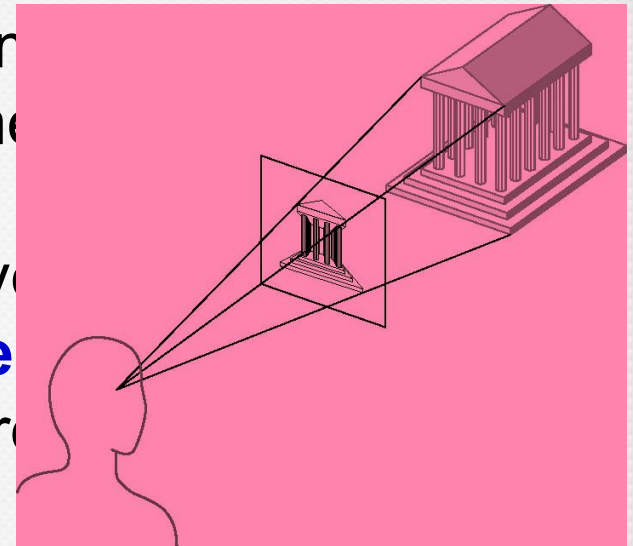
Perspective Viewing

Perspective Viewing

Unit 4

Viewing

- All perspective projectors are characterized by **diminution of size**.
- When objects are moved farther from the viewer, their images become smaller.
- This size change gives perspective projection its **natural appearance**. Hence it is widely used in architecture and animation.
- Projectors converge at COP.



Perspective Viewing

Unit 4

Viewing



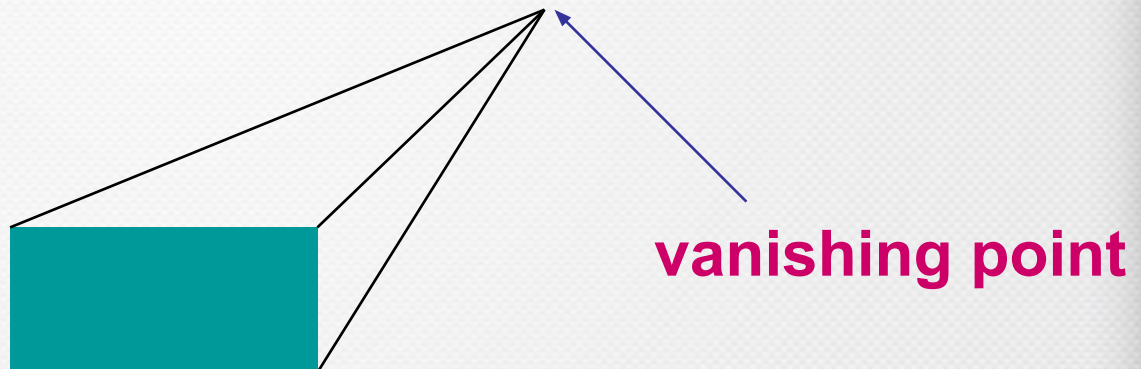
Perspective Viewing

Unit 4

Viewing

Vanishing Points

- Parallel lines (not parallel to the projection plane) on the object converge at a single point in the projection (the *vanishing point*).
- Drawing simple perspectives by hand uses these vanishing point(s).



Unit 4

Viewing

Types of Perspective Viewing

Perspective Viewing

Unit 4

Viewing

Types of perspective views

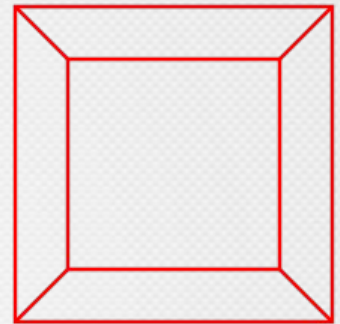
- Based upon how many of the three principal directions in the object are parallel to the projection plane.
- The names of these categories refer to the number of *vanishing points* in the perspective drawing.
 - One point perspective
 - Two point perspective
 - Three point perspective

1 Point Perspective Viewing

Unit 4

Viewing

- Two of the principal directions are parallel to the projection plane.
- **Single vanishing point.**
- Used for roads, railway tracks, hallways, or buildings viewed so that the front is directly facing the viewer.



1 Point Perspective Viewing

Unit 4

Viewing

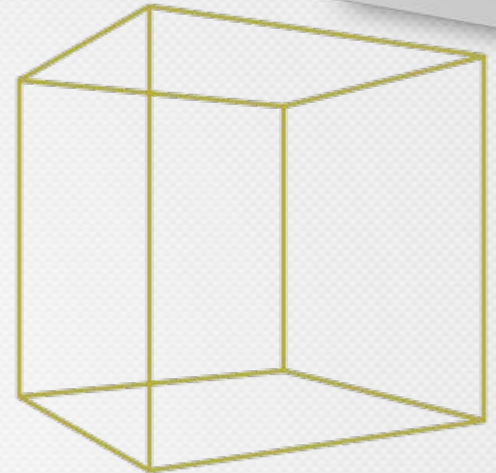


2 Point Perspective Viewing

Unit 4

Viewing

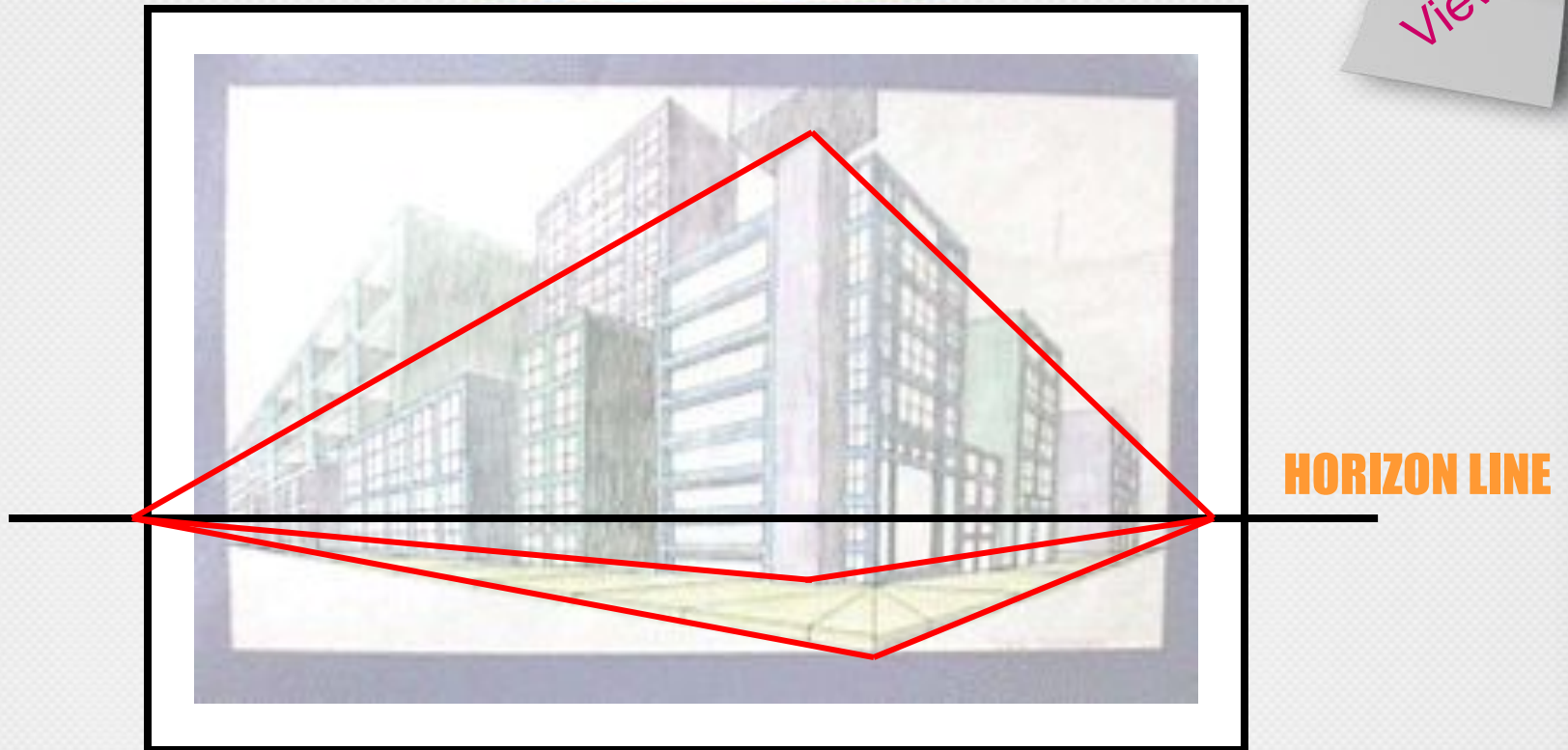
- **One of the principal directions** is parallel to projection plane.
- **Two vanishing points.** (Lines in only two of the principal directions converge at vanishing points)



2 Point Perspective Viewing

Unit 4

Viewing

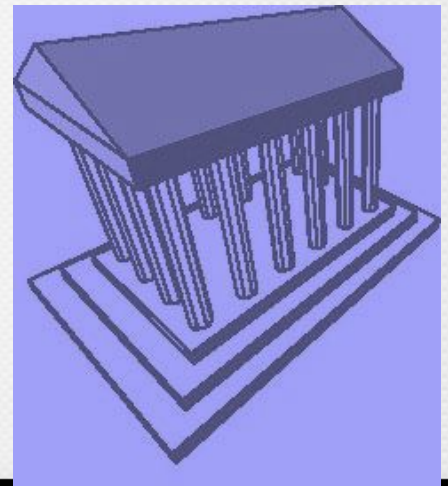
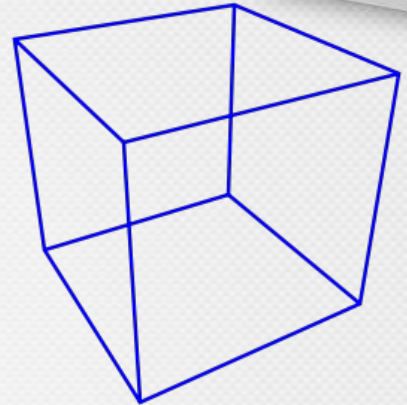


3 Point Perspective Viewing

Unit 4

Viewing

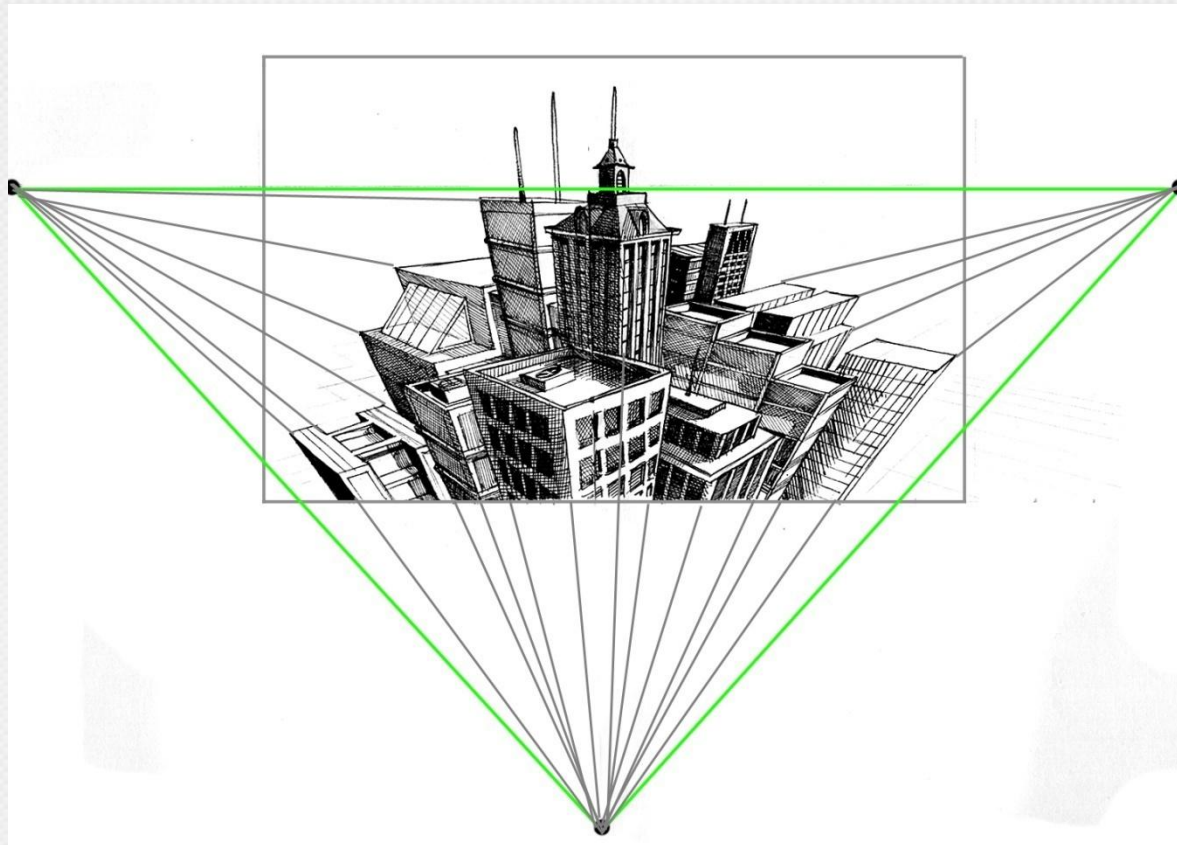
- **No principal direction** is parallel to projection plane.
- **3 vanishing points.** (The parallel lines in each of the three principal directions converge at vanishing point).



3 Point Perspective Viewing

Unit 4

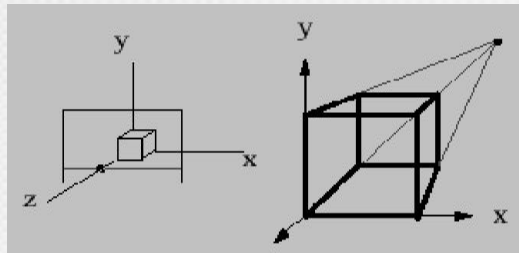
Viewing



Types of Perspective Projection

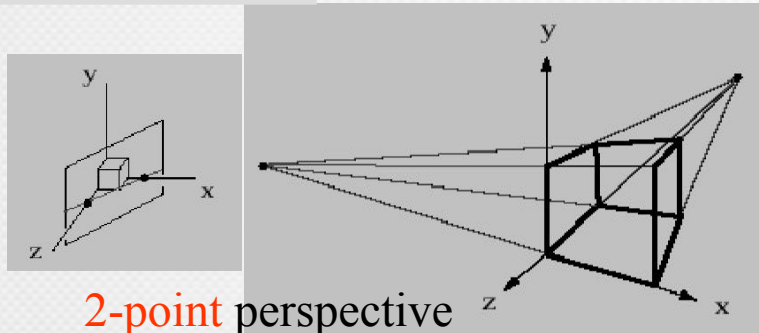
Unit 4

Viewing

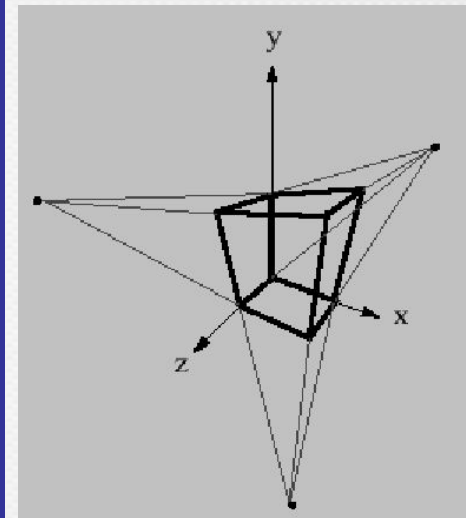
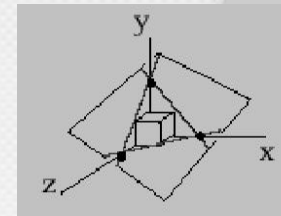


1-point perspective

Most popular



2-point perspective



3-point perspective

Unit 4

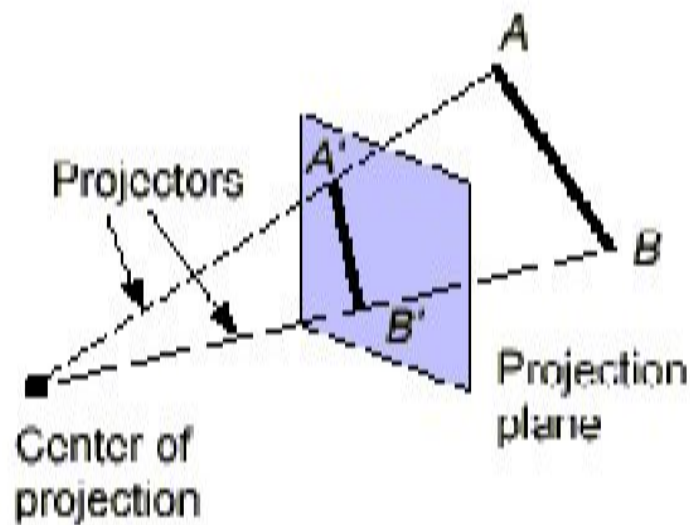
Viewing

Differences between parallel and perspective projection

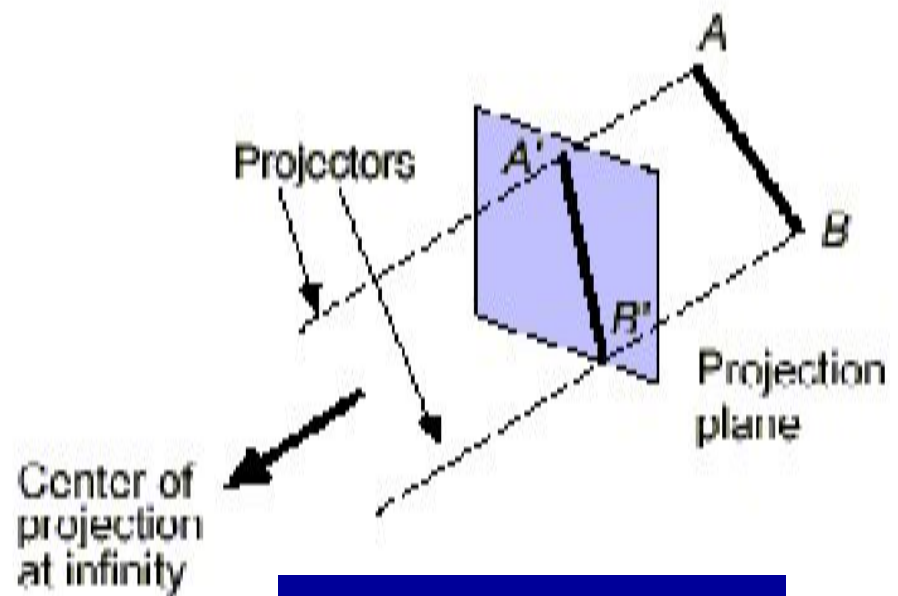
Differences between Parallel and Perspective projection

Unit 4

Viewing



Perspective Projection



Parallel Projection

Perspective Projections

Unit 4

Viewing

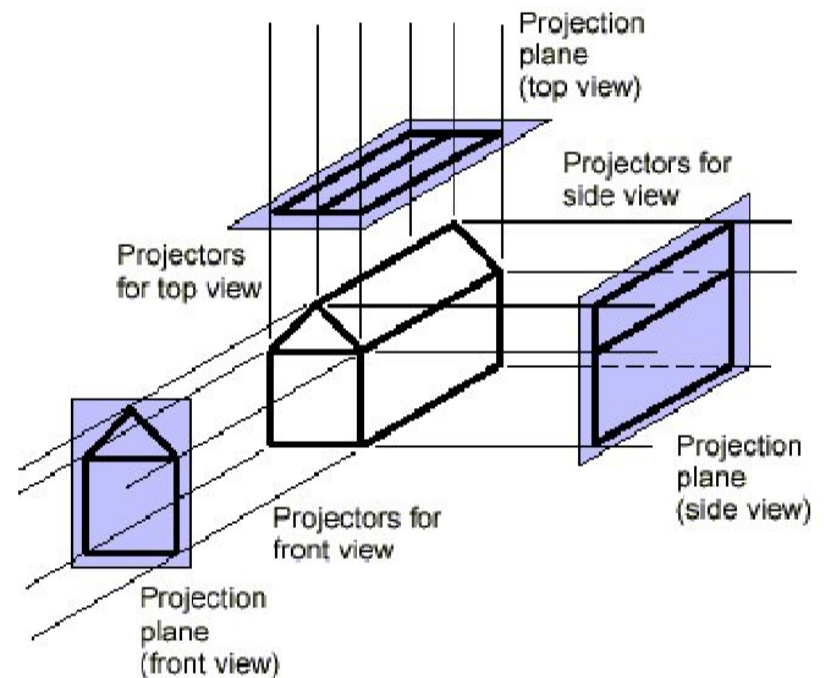
- Projectors are not parallel to one another.
- Give a **realistic** view of 3D objects, but does not preserve shape of object or scale.
- Perspective images of parallel lines not parallel with projection plane converge to a point called **vanishing point**.
- Size of object is diminished with distance.
- Used in architectural, engineering, industrial design, and advertising drawings.

Parallel Projections

Unit 4

Viewing

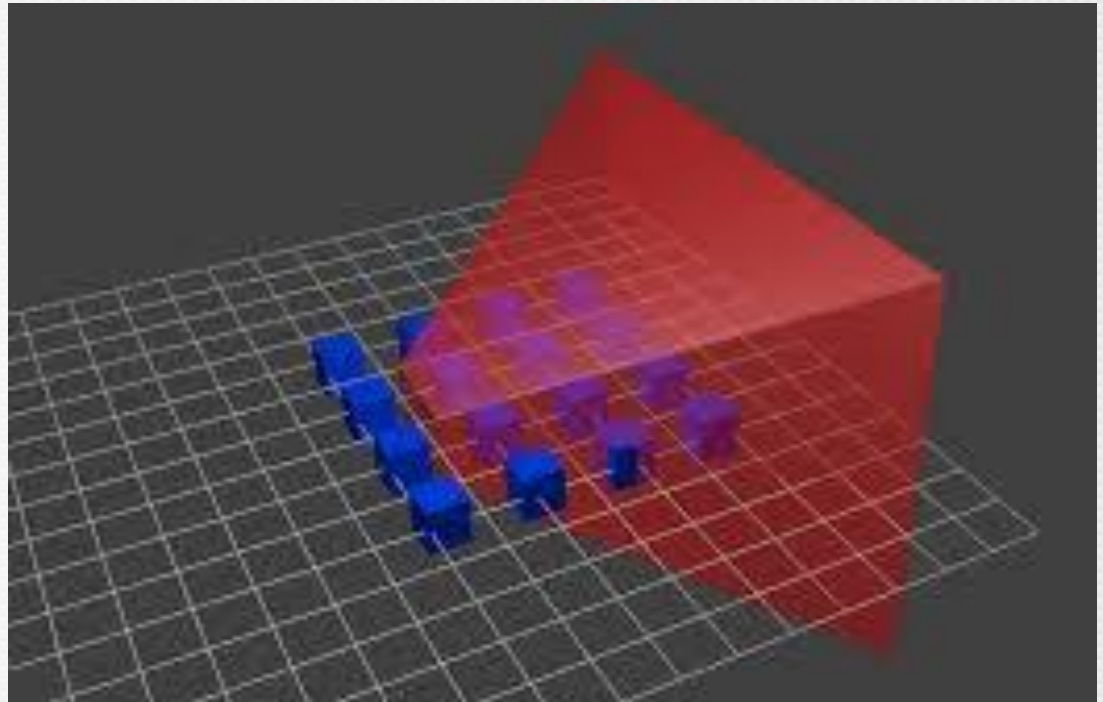
- **Accurate** and **same-scale** measurements.
- **Not realistic**, and may **require multiple projections** to get a complete feeling of the object.
- Used in drawings of machine parts and in working architectural drawings.
- All projectors are parallel to one another because camera is placed at infinite distance.



Unit 4

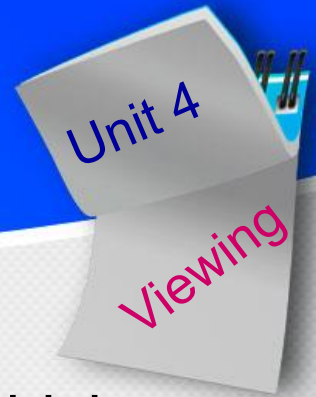
Viewing

Computer View



OpenG

Viewing with a Computer



There are 3 aspects of the viewing process, all of which are implemented in the pipeline

- **Positioning and orientation of the camera**
Setting the model view matrix
- **Selecting a lens**
Setting the projection matrix
- **Clipping**
Setting the view volume

Viewing with a Computer

Unit 4

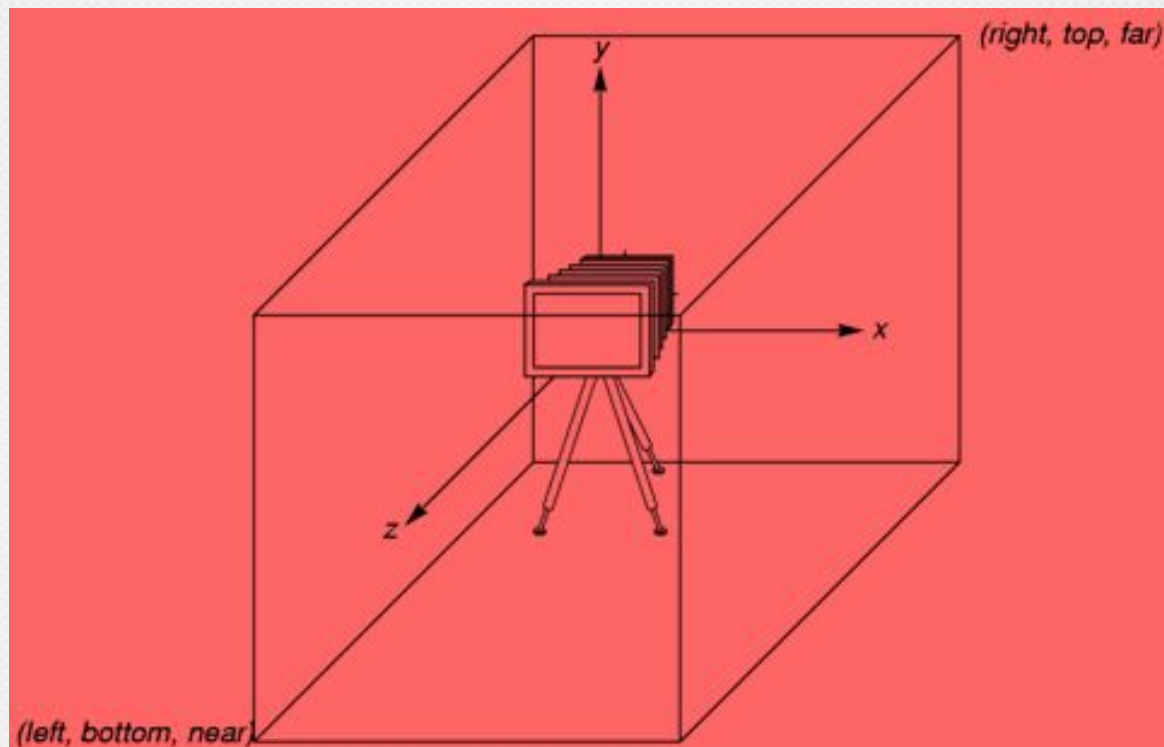
Viewing

- OpenGL starts with the camera at the origin of the object frame pointing in **the -ve z direction**.
- The camera is set up for the **orthogonal views** and has a viewing volume , a cube centred at the origin with sides of length 2.
- Default projection plane is the plane $z=0$.
- Direction of the projection is along z axis.
- Both **model-view** and **projection matrices** are left as default **identity matrices**.
- To place the camera at the desired position, change model-view matrix.

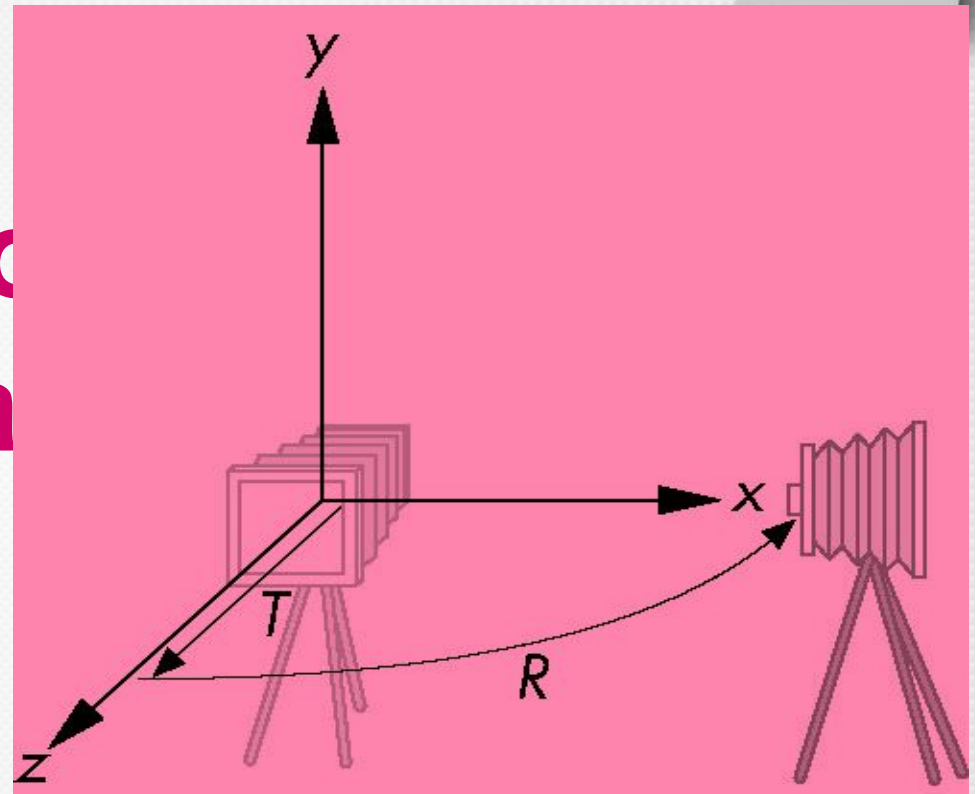
Viewing with a Computer

Unit 4

Viewing



Positioning of the Camera

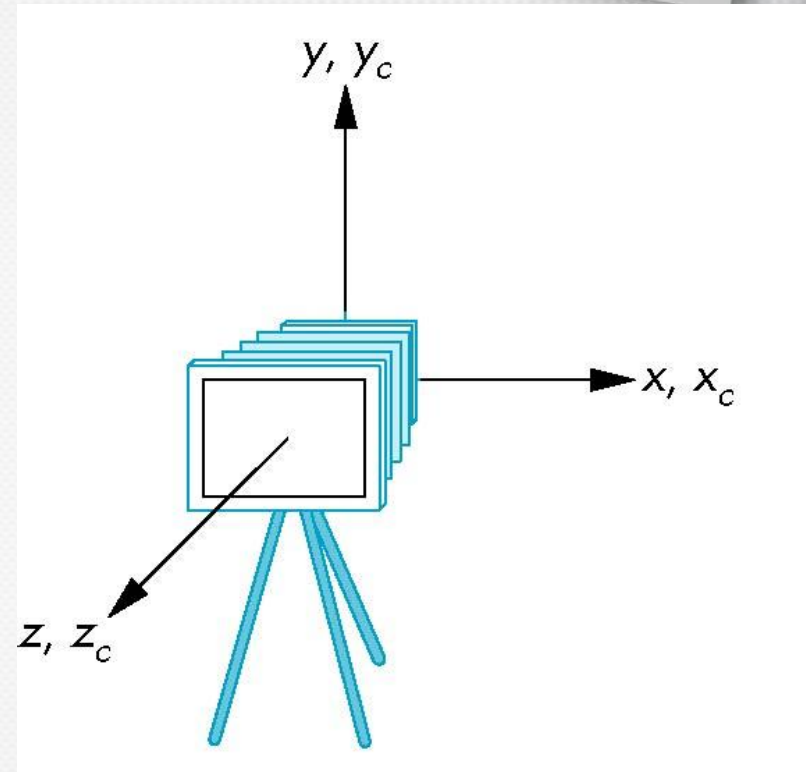


Positioning of the camera

Unit 4

Viewing

- In OpenGL, initially the object and camera frames are the same.
- *Default model-view matrix* is an *identity*.
- The camera is located at origin and points *in the negative z direction*.

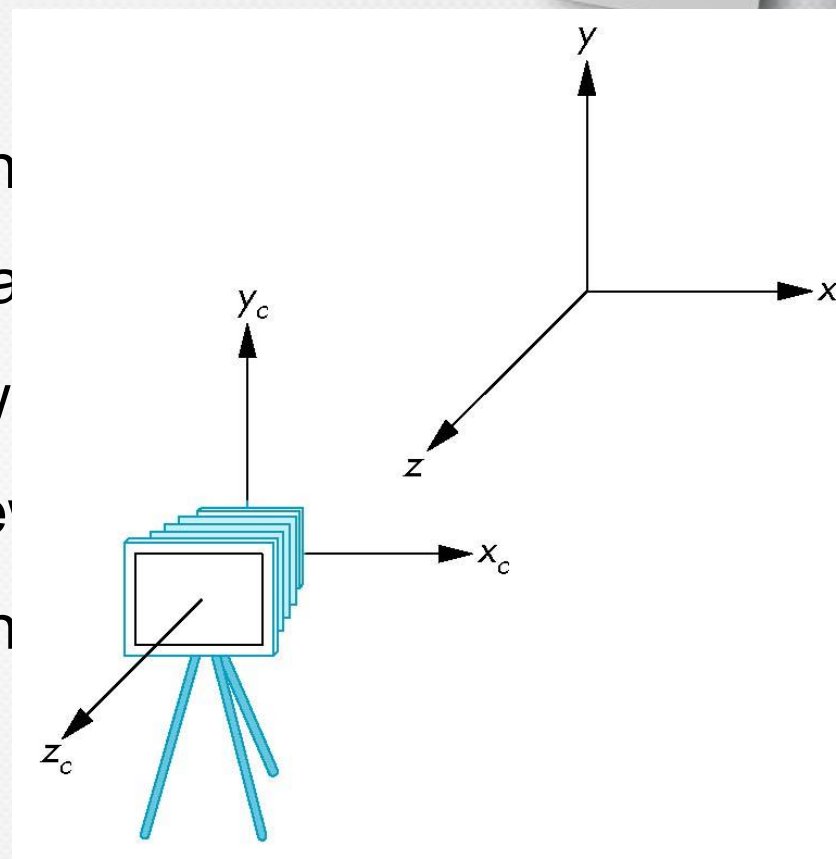


Positioning of the camera

Unit 4

Viewing

- **Default projection matrix is an identity.**
- With the default position, all the objects present in the scene may not be visible and hence we will have to change the model view matrix suitably to position the camera at the desired location.

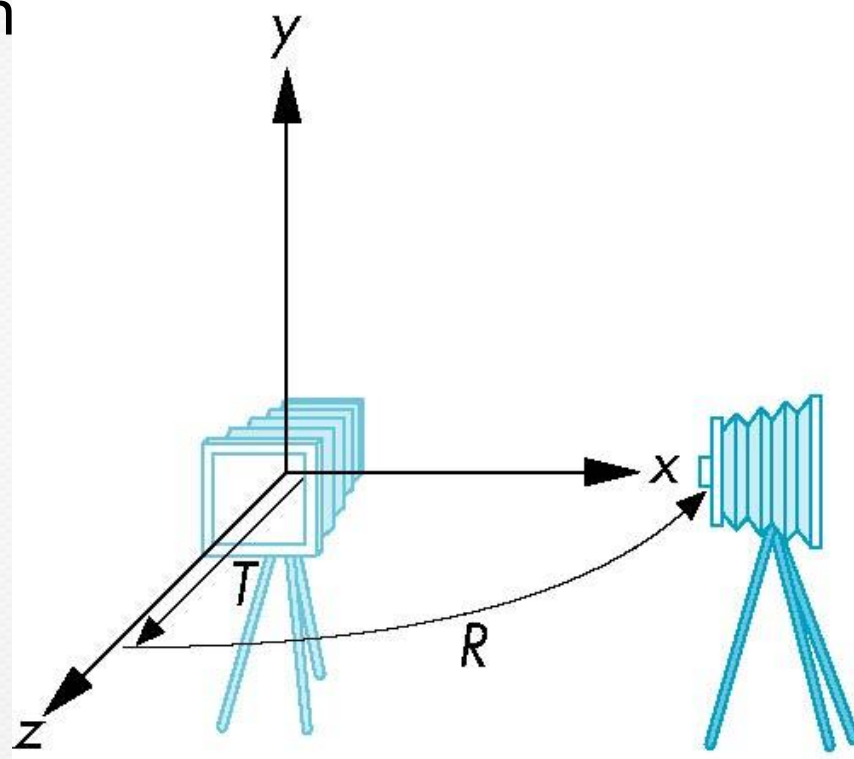


Positioning of the camera

Unit 4

Viewing

- We can move the camera to any desired position by a sequence of rotations and translations
- **Example: side view**
 - **Rotate** the camera
 - **Move it** away from origin
 - Model-view matrix $C = TR$



Positioning of the camera

Unit 4

Viewing

Example1

- Consider an object centred at the origin.
- The camera is also in its initial position (at the origin, pointing down $-ve$ z -axis).
- Suppose the user wants to see an image of the faces of the object that point in **+ve z direction**, we must move the camera away from the origin.

Positioning of the camera

Unit 4

Viewing

Example1

- If we allow the camera to remain pointing in the -ve z direction, then we want to move the camera backward along +ve z-axis. The camera can be positioned using

```
glMatrixMode(GL_PROJECTION);
```

```
glLoadIdentity();
```

```
glTranslatef(0.0, 0.0, -d);
```

Positioning of the camera

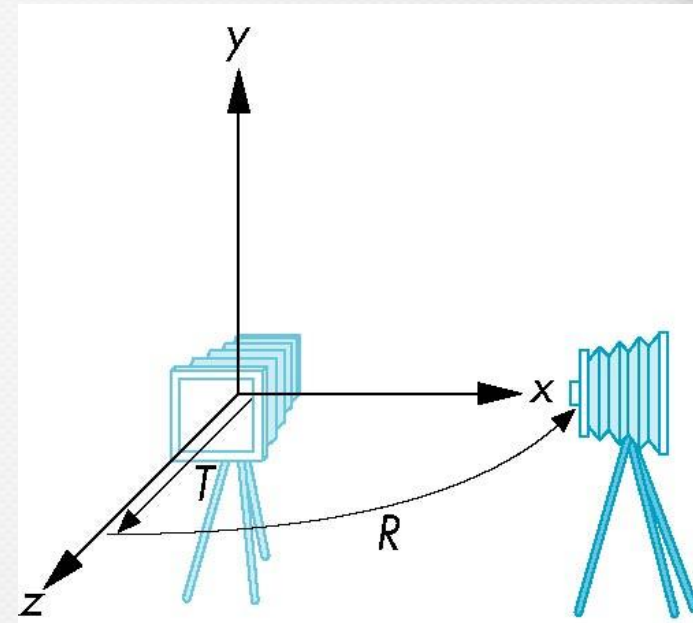
Unit 4

Viewing

- Example2

Suppose the user is interested in viewing at the same object from **+ve x axis**. Then the camera can be positioned using

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(0.0, 0.0, -d);  
glRotatef(-90.0, 0.0, 1.0, 0.0);
```



Viewing APIs

Unit 4

Viewing

We also have OpenGL functions for the normalization transformation.

Normalization transformation is a process of viewing (parallel or perspective).

1. **View Reference Point (VRP)** : to specify the position of the camera.

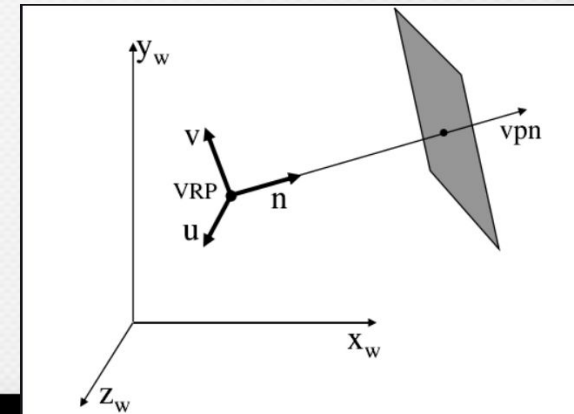
set_view_reference(x,y,z);

2. **View Plane Normal (VPN)** : to specify the orientation of the projection plane.

set_view_plane_normal(nx, ny,nz);

3. **View Up Vector (VUP)** : to fix the camera.

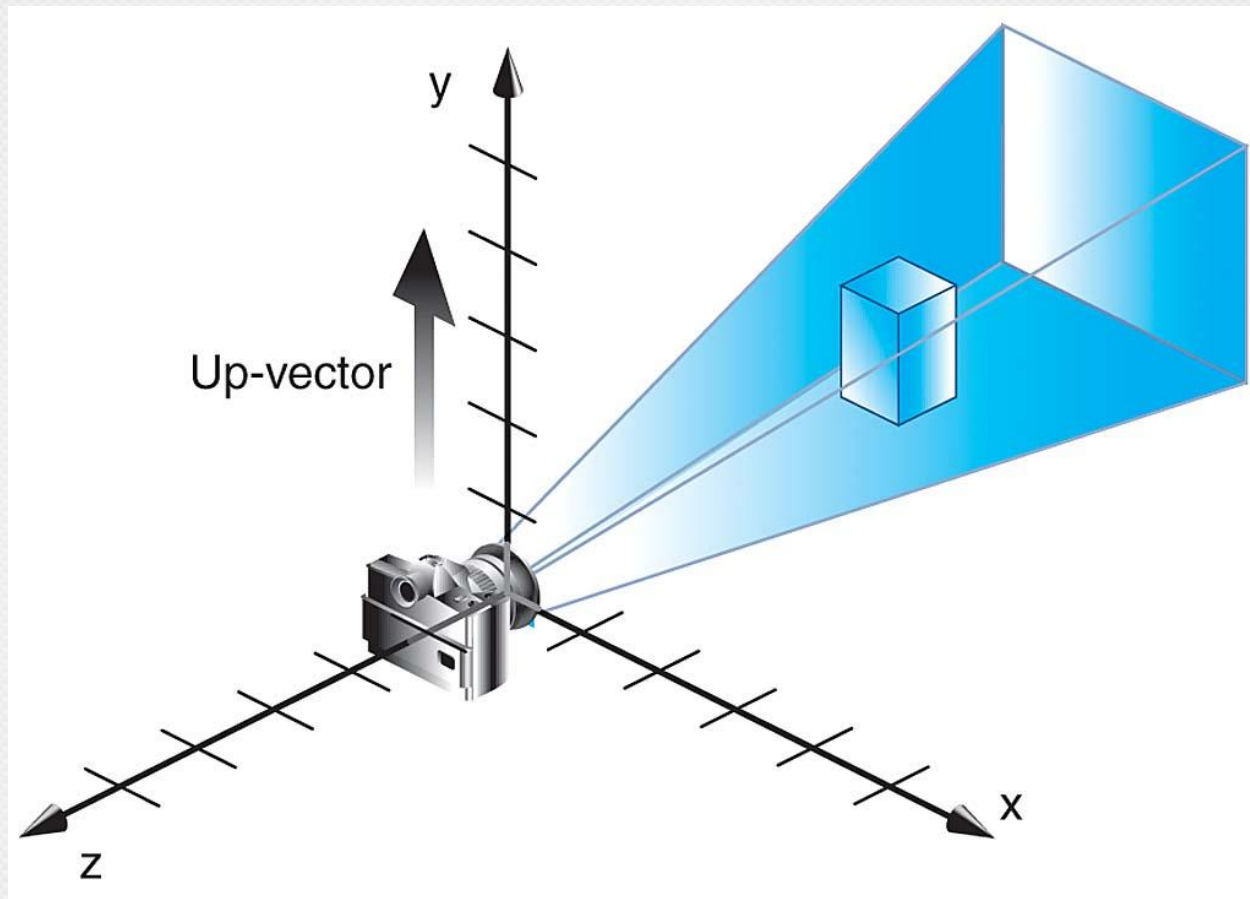
set_view_up(vup_x, vup_y,v_up_z);



Look_At Function

Unit 4

Viewing



Look_At Function

Unit 4

Viewing

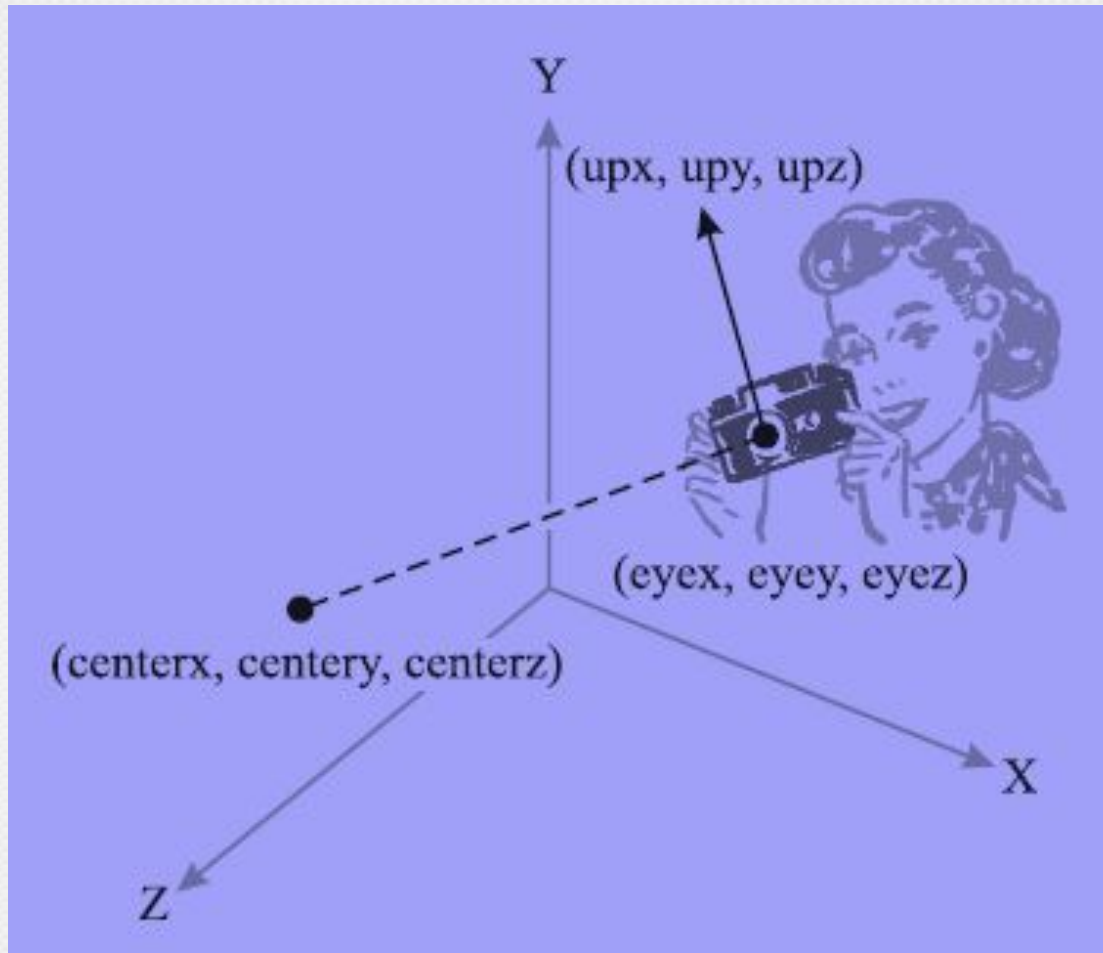
```
void gluLookAt(GLdouble eyex, GLdouble eyey, GLdouble eyez,  
               GLdouble atx, GLdouble aty, GLdouble atz,  
               GLdouble upx, GLdouble upy, GLdouble upz);
```

```
glLoadIdentity ();           // start with a unit modelview matrix  
gluLookAt  
(0.0, 0.0, 0.5,             // position of camera  
 0.0, 0.0, 0.0,             // point at which camera looks  
 0.0, 1.0, 0.0);           // up direction of camera
```

Look_At Function

Unit 4

Viewing



```
void gluLookAt(GLdouble eyex, GLdouble eyey, GLdouble eyez,  
               GLdouble atx, GLdouble aty, GLdouble atz,  
               GLdouble upx, GLdouble upy, GLdouble upz);
```

Unit 4

Viewing

- The **gluLookAt** function creates a **viewing matrix** derived from an eye point, a reference point indicating the center of the scene, and an up vector.
- The matrix maps
 - the **eye point** to the **origin**.
 - **reference (at) point** to the **-ve z-axis**.
 - **up vector** direction to the **+ve y-axis** so that it points upward in the viewport.
- The up vector must not be parallel to the line of sight from the eye to the reference point.

Perspective Projections

Unit 4

Viewing

- Suppose we are in the camera frame with the camera located at the origin pointed in the -ve z direction.
- Two possibilities
 - **Back of the camera parallel to front face**
The back of the camera is orthogonal to z direction and is parallel to lens.
 - **Back of the camera not parallel to front face**
The back of the camera can have any orientation with respect to front face.

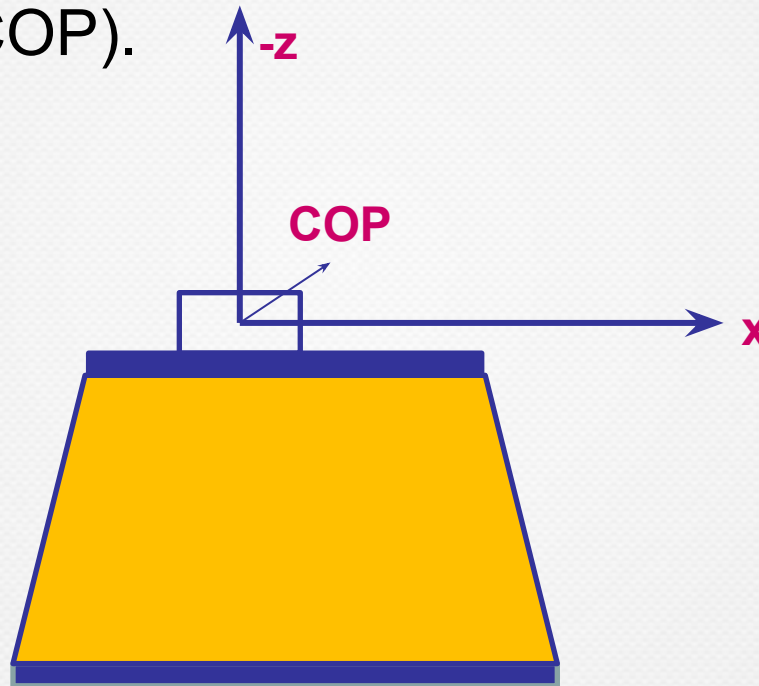
Perspective Projections

Unit 4

Viewing

Back of the camera parallel to front face

We can place the projection plane in front of the Centre Of Projection(COP).



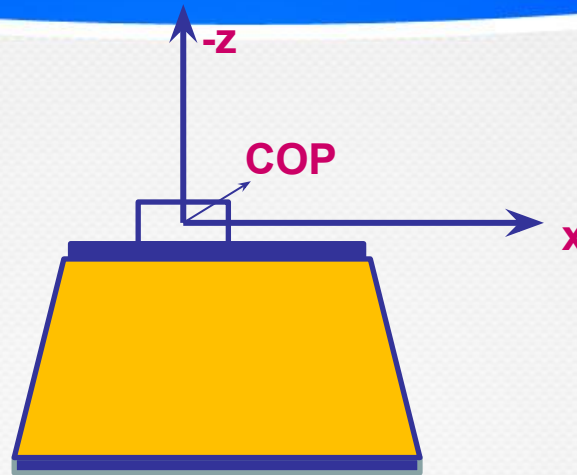
Perspective Projections

Back of the camera parallel to front face

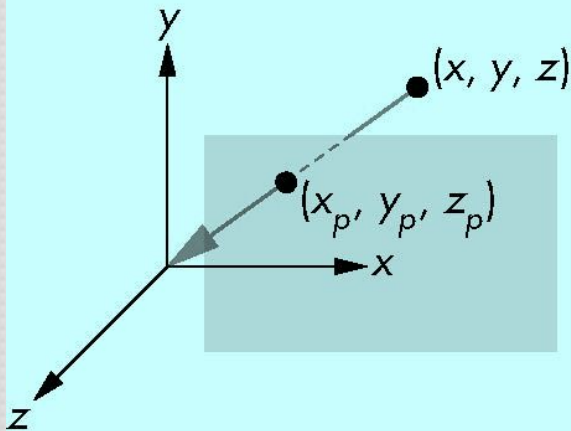
Unit 4

Viewing

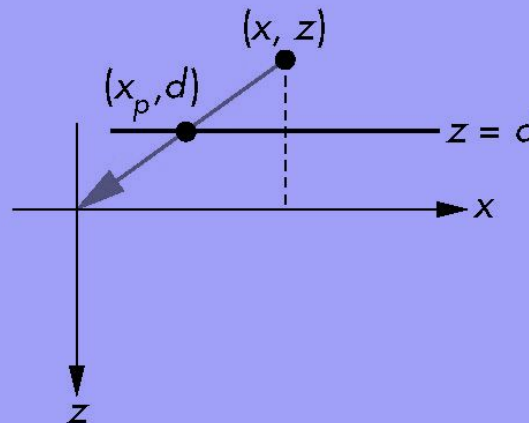
Orthogonal
projection



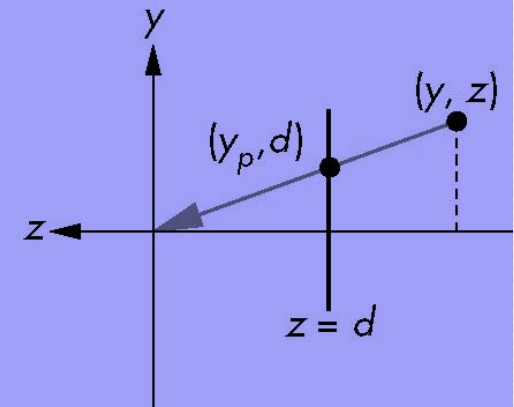
3D view



Top view



Side view



Perspective Projections

Back of the camera parallel to front face

Unit 4

Viewing

- A point in space (x, y, z) is projected along a projector into the point (x_p, y_p, z_p) . All projectors pass through the origin.
- Because the projection plane is perpendicular to the z axis, $z_p = d$.
- Because the camera is pointing in the -ve z direction, the value of d is -ve.

Perspective Projections

Back of the camera parallel to front face

Unit 4

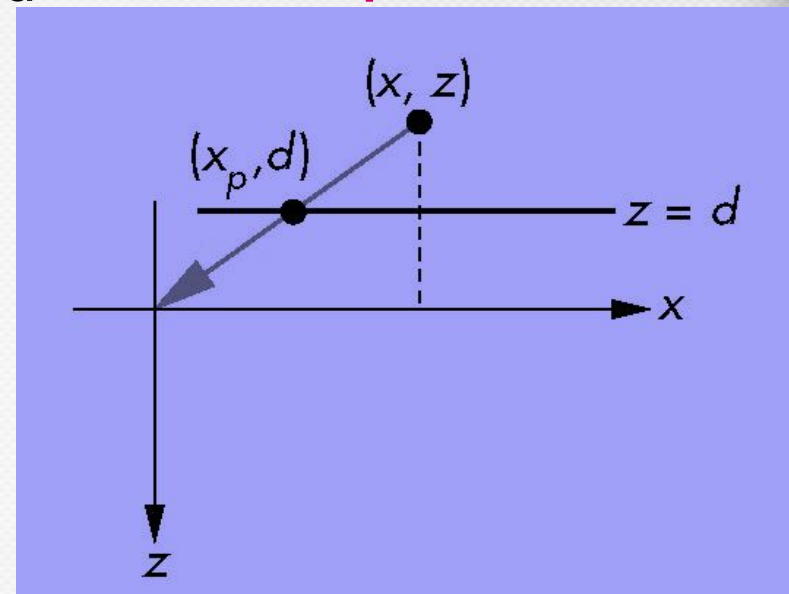
Viewing

Top View

- Two triangles are similar and tangents are same.
- $x/z = x_p/d$

$$x_p = \frac{x}{z/d}$$

Top view



Perspective Projections

Back of the camera parallel to front face

Unit 4

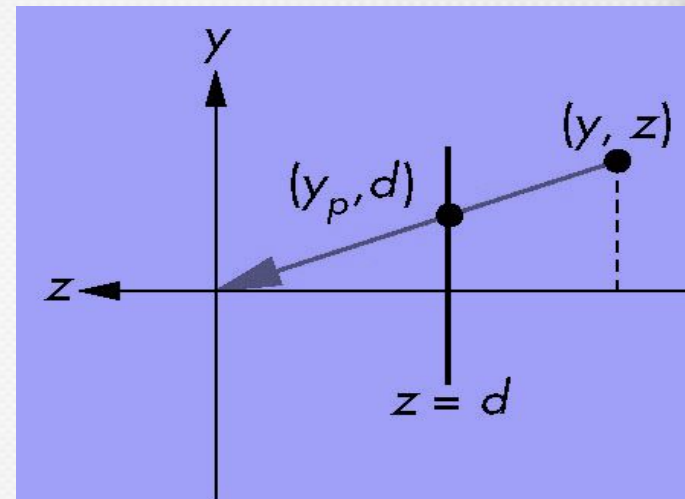
Viewing

Side View

- Two triangles are similar and tangents are same.
- $y/z = y_p/d$

$$y_p = \frac{y}{z/d}$$

Side view



Perspective Projections

Back of the camera parallel to front face

Unit 4

Viewing

$$x_p = \frac{x}{z/d}$$

$$y_p = \frac{y}{z/d}$$

- The equations are nonlinear.
- The division by z indicates that the image of the object farther from COP is smaller compared to image of the object closer to COP.
- The equations can be obtained in the matrix form.

Perspective Projections

Back of the camera parallel to front face

Unit 4

Viewing

Consider the matrix

The matrix M transforms the point

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/d & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} \text{ to the point } \begin{pmatrix} x_p \\ y_p \\ z_p \\ z/d \end{pmatrix}$$

$$x_p = \frac{x}{z/d}$$

$$y_p = \frac{y}{z/d}$$

$$z_p = d$$

Perspective Projections

Back of the camera parallel to front face

Unit 4

Viewing

$$\begin{pmatrix} x_p \\ y_p \\ z_p \\ z/d \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/d & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Perspective Projections

Back of the camera parallel to front face

Unit 4

Viewing

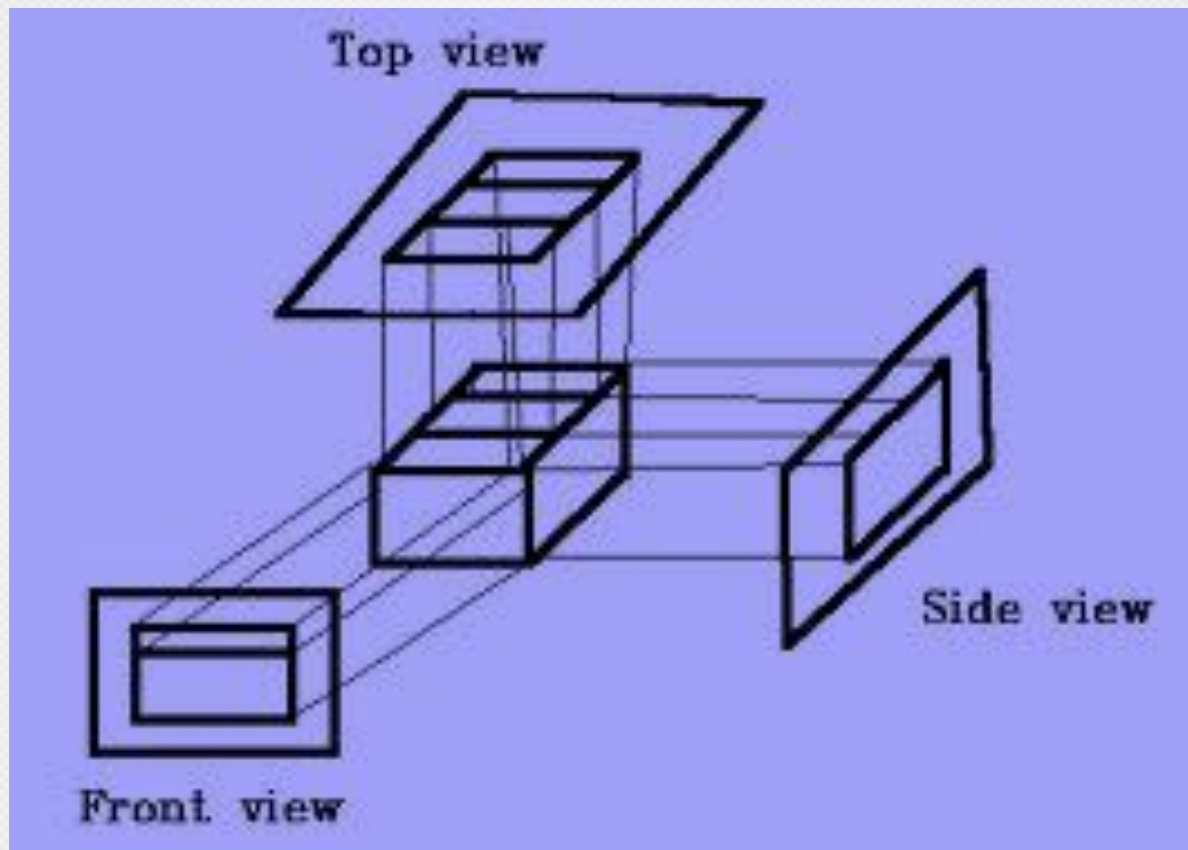
By dividing 1st 3 terms by the 4th term,

$$q' = \begin{pmatrix} x_p \\ y_p \\ z_p \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{x}{z/d} \\ \frac{y}{z/d} \\ d \\ 1 \end{pmatrix}$$

Orthogonal Projections

Unit 4

Viewing

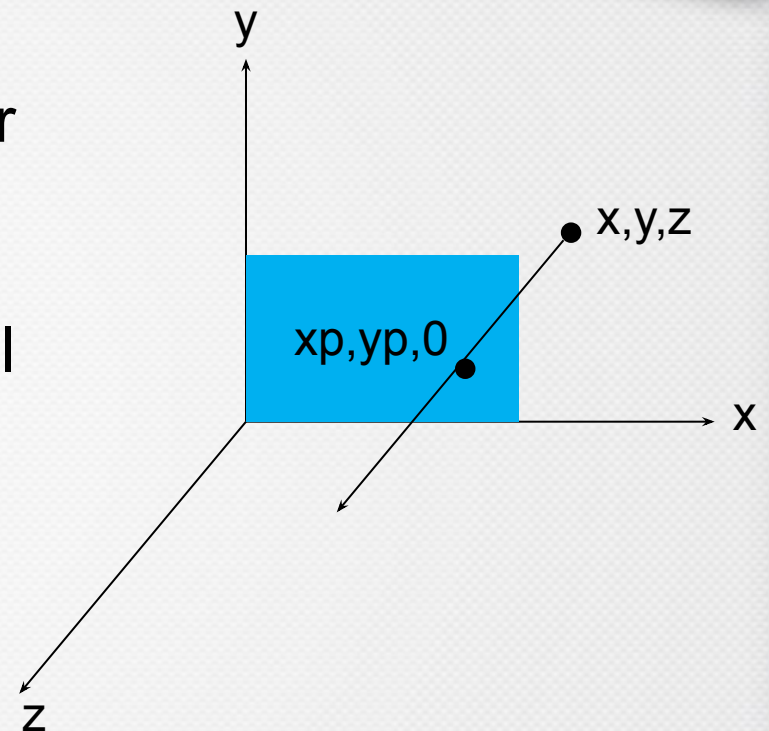


Orthogonal Projections

Unit 4

Viewing

- The projectors are perpendicular to the view plane.
- Camera lens has infinite focal length.



Perspective
projection

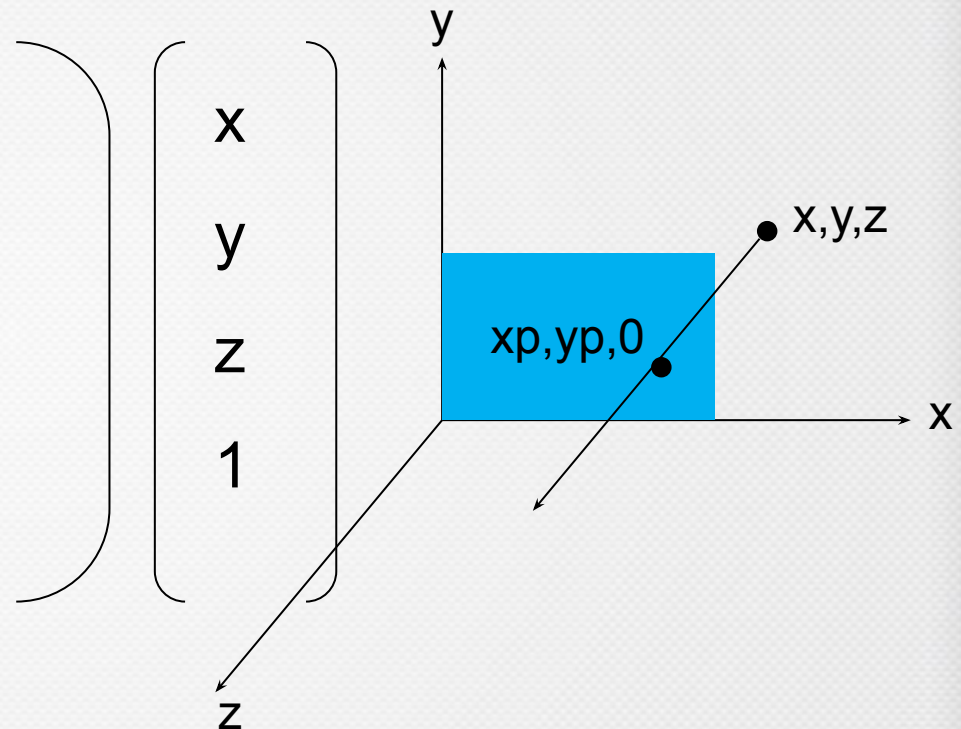
Orthogonal Projections

Unit 4

Viewing

$$x_p = x, y_p = y, z_p = 0$$

$$\begin{pmatrix} x_p \\ y_p \\ z_p \\ 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$



Projections in OpenGL

Unit 4

Viewing

1. Perspective Viewing in OpenGL

`glFrustum()`

`gluPerspective()`

2. Parallel Viewing in OpenGL

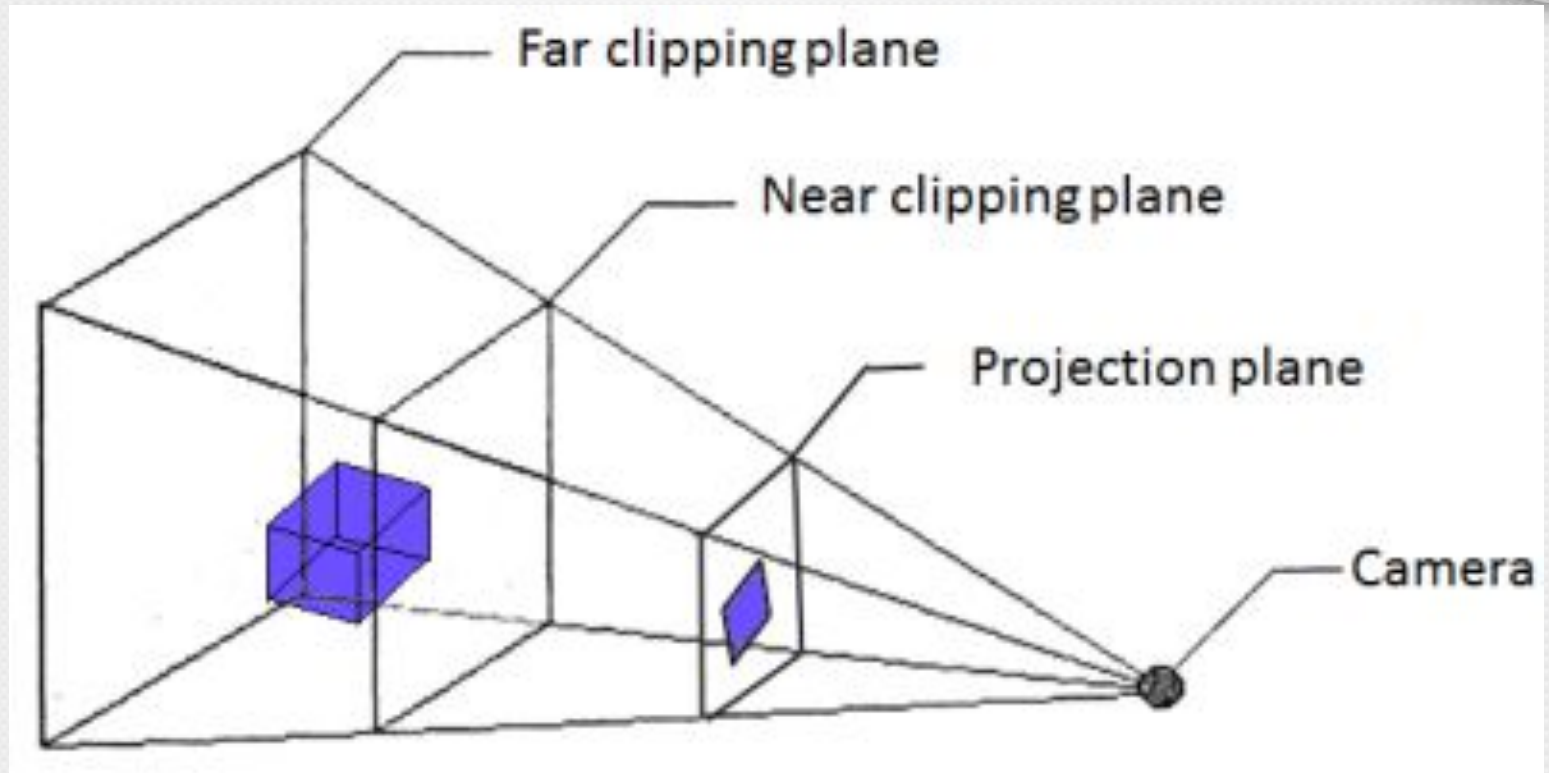
`glOrtho()`

`gluOrtho2D()`

Perspective Projections in OpenGL

Unit 4

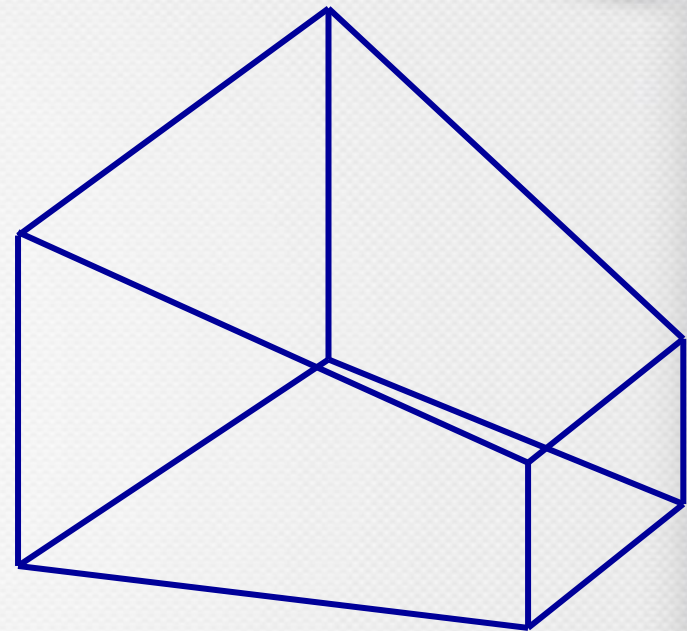
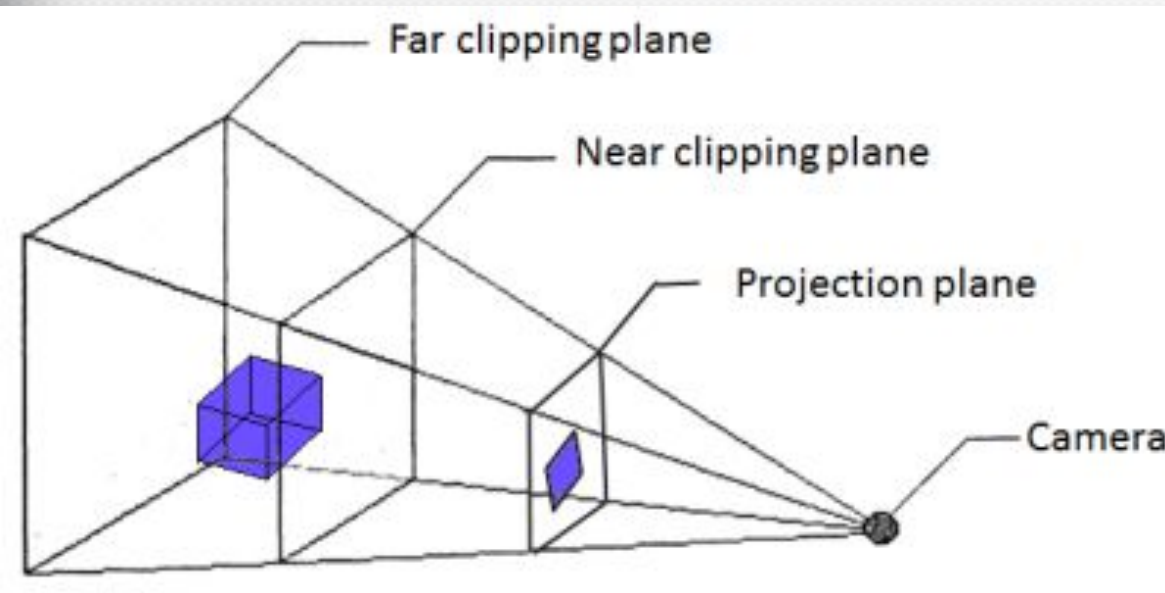
Viewing



Perspective Projections in OpenGL

Unit 4

Viewing



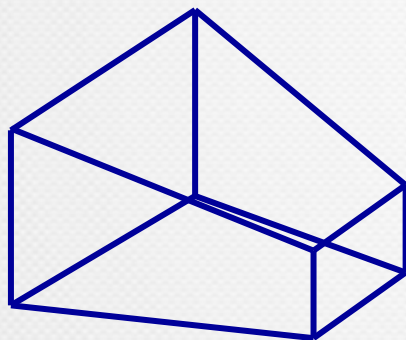
Frustum View volume

Perspective Projections in OpenGL

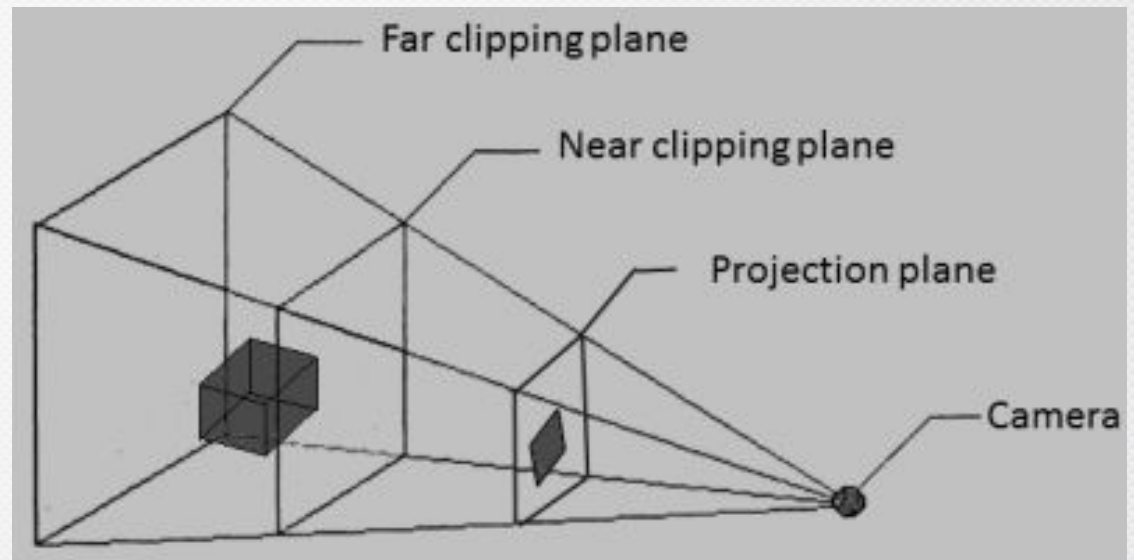
Unit 4

Viewing

- Objects outside the view volume not displayed.
- View volume has two planes.
 - The front face is called near plane.
 - The back face is called far plane.
- All vertical planes from near plane to far plane define the clipping area.



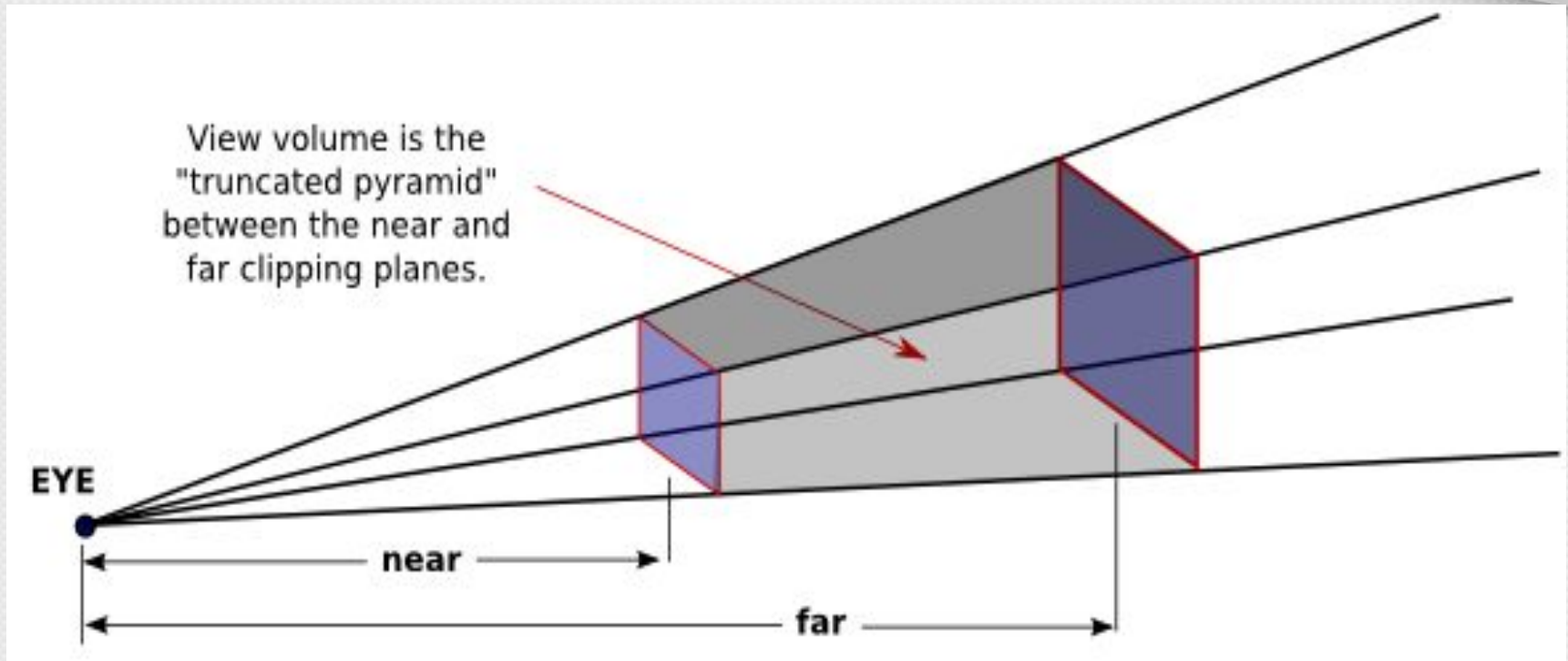
Frustum View
volume



Perspective Projections in OpenGL

Unit 4

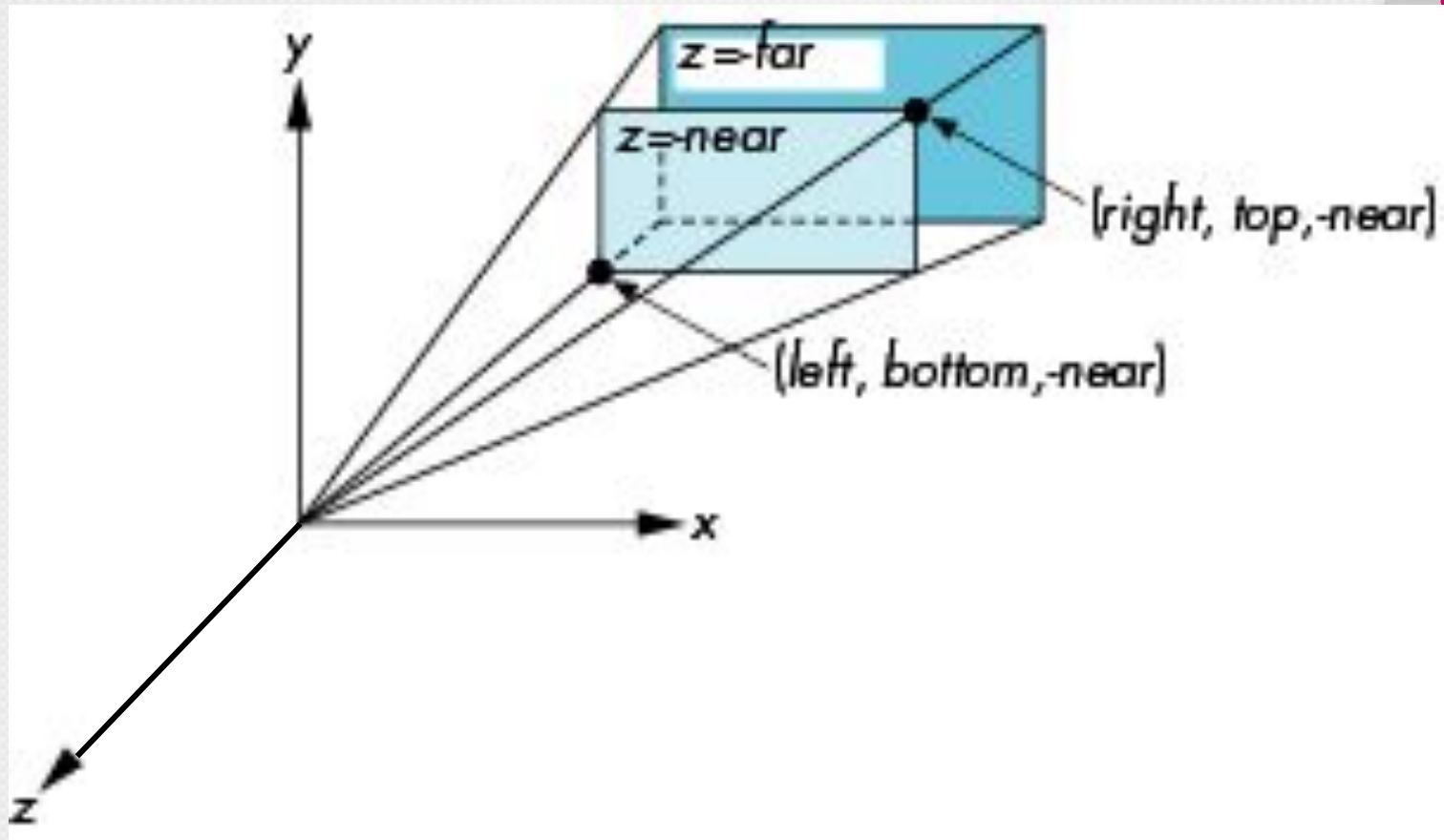
Viewing



Perspective Projections in OpenGL

Unit 4

Viewing



Perspective Projections in OpenGL

Unit 4

Viewing

```
void glFrustum( Gldouble left, Gldouble right,  
                Gldouble bottom, Gldouble top,  
                Gldouble near, Gldouble far)
```

glFrustum computes a viewing frustum in the world coordinates.

Perspective Projections in OpenGL

Unit 4

Viewing

- **left and right**
 - Specify the coordinates for the left and right vertical clipping planes.
- **bottom and top**
 - Specify the coordinates for the bottom and top horizontal clipping planes.
- **near and far**
 - Measured from the COP (origin in eye coordinates) to front and back clipping planes.
 - Specify the distances to the near and far depth clipping planes. **Both distances must be positive.**

Perspective Projections in OpenGL

Unit 4

Viewing

```
glMatrixMode(GL_PROJECTION);
```

```
glLoadIdentity();
```

```
glFrustum(left, right, bottom, top, near, far);
```

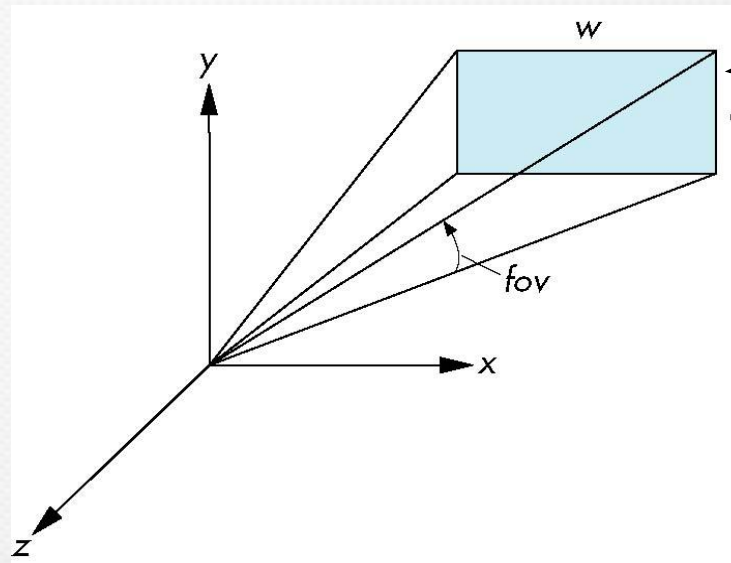
Perspective Projections in OpenGL

Using Field of View

Unit 4

Viewing

- With **glFrustum** it is often difficult to get the desired view.
- **gluPerspective(fovy, aspect, near, far)** often provides a better interface.



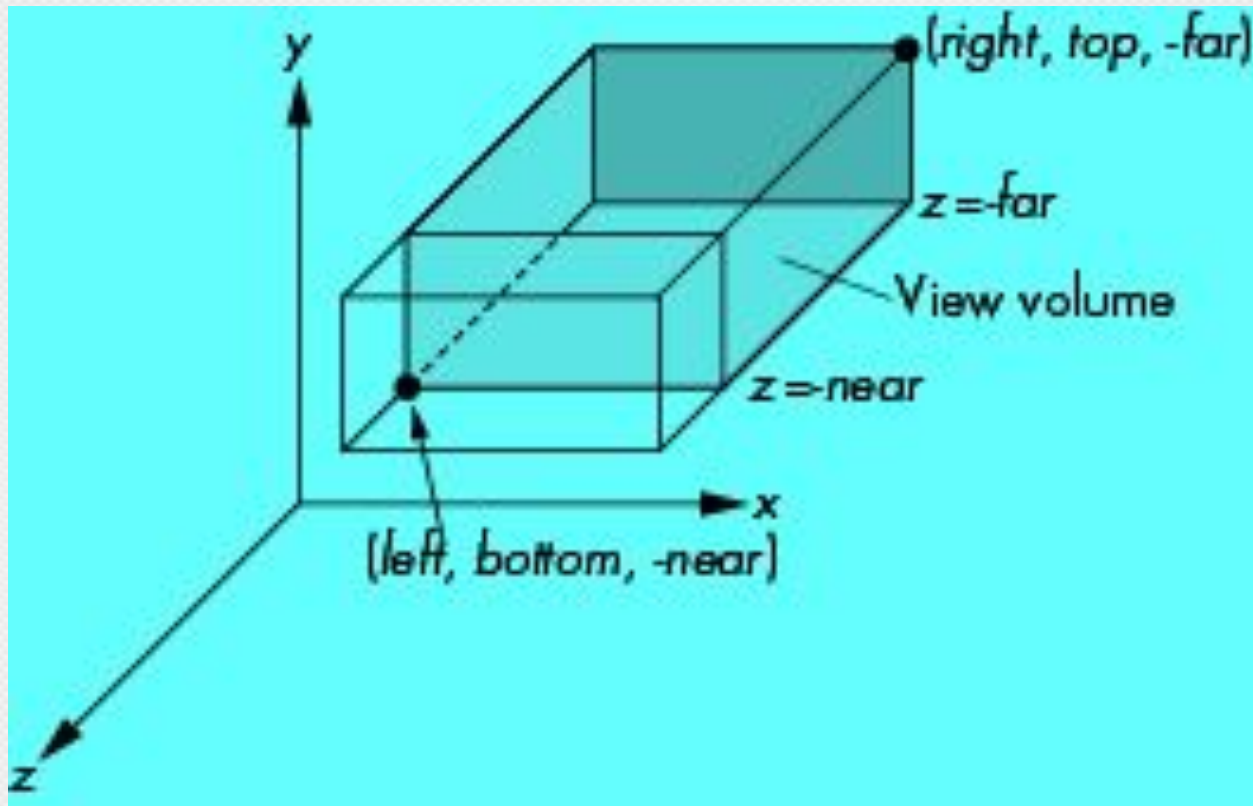
front plane

$$\text{aspect} = w/h$$

Parallel Projections in OpenGL

Unit 4

Viewing

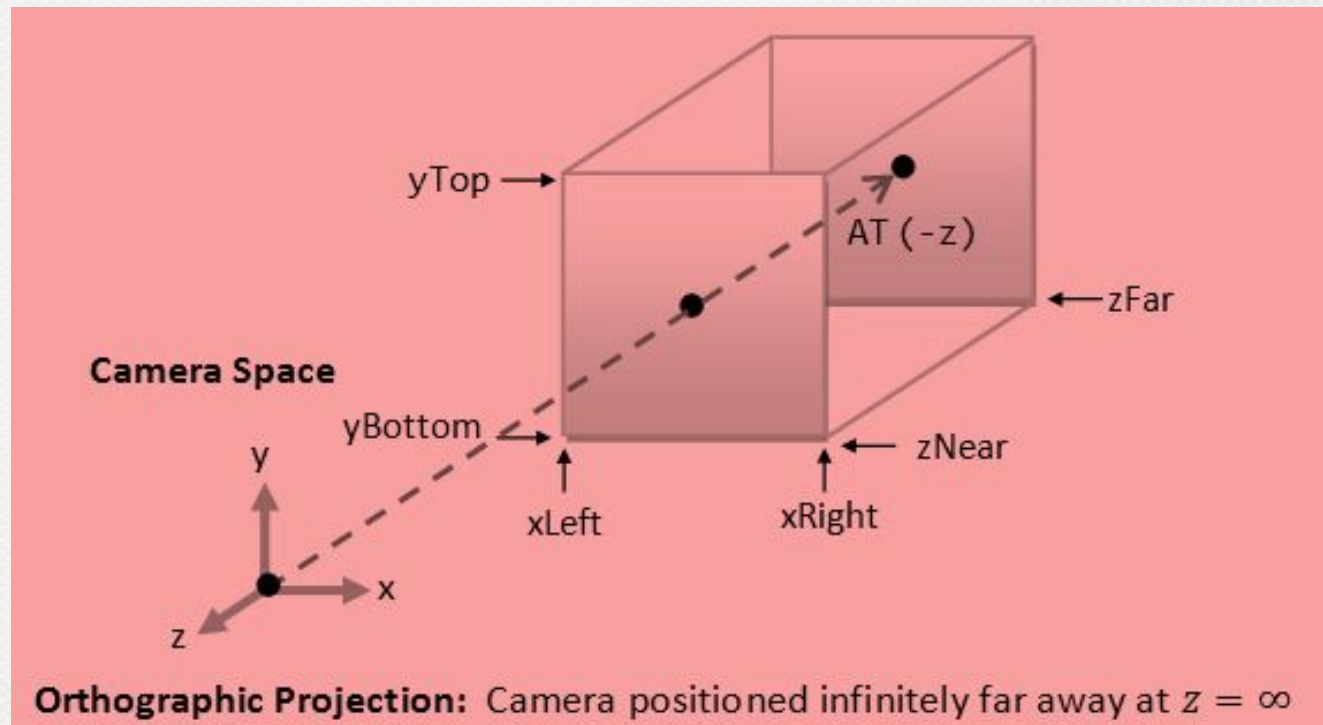


Parallel Projections in OpenGL

Unit 4

Viewing

`glOrtho(GLdouble left, GLdouble right, GLdouble bottom, GLdouble top, GLdouble near, GLdouble far);`



Oblique Projection



Cavalier:

No foreshortening of lines which are perpendicular to PP

$$\tan \theta = 1, \cot \theta = 1$$

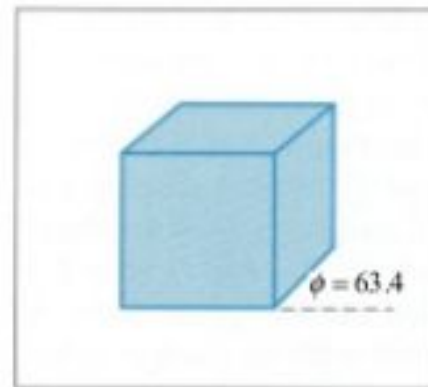
Cabinet:

Lines which are perpendicular to PP are foreshortened by half

$$\tan \theta = \tan 63.4 = 2, \cot \theta = 1/2$$



Cavalier
(DOP $\theta = 45^\circ$)



Cabinet
(DOP $\theta = 63.4^\circ$)

Viewing Pipeline Revisited

